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CivicWorkOS: Capability-Preserving Allocation of Municipal Work Among Humans, AI Agents, and Robots

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ABSTRACT

Cities are moving from sensing infrastructure to agentic operation, in which AI agents perform administrative work and robots perform physical work. Allocation research optimizes fit, throughput, and ergonomics; labor research examines displacement; workforce rostering allocates training hours once the human-versus-machine decision has been made. None of them weighs, in that prior decision, the developmental practice through which cities form their future practitioners. We present CivicWorkOS, which makes municipal allocation a governed decision over *staffed modes* (pairings of an execution mode with a named roster at declared career stages) rather than a human-versus-machine choice. Civic Automation Debt penalizes five deferred liabilities, and four constraints restrict the allocation: a Human Capability Preservation Budget denominated in qualified-practice hours accruing only to practitioners below independent competence, a Capability Access Constraint governing who receives protected practice, a Just Transition Constraint, and an $N-1/N-2$ resilience reserve. We formulate the city-wide program, an online Lagrangian rule with a bounded between-rebalance violation, a necessary feasibility condition, and a trainee-cohort requirement giving the steady-state number of developing practitioners the capability pipeline needs. In an exact bridge-inspection instance with illustrative inputs, reproducible by direct arithmetic from the reported values, the capability budget binds with a marginal opportunity cost of 0.0454 objective units per protected-practice hour and costs 4.34% of annual objective value (0.33–9.60% across the swept parameter values); meeting it at the chosen replacement factor of 1.2 forms 3.46 competent-equivalents a year against attrition of 2.88. In this domain, machine assistance (a factor of 1.69) and sharing the work with a supervising lead (a factor of 2.0) together stretch time to competence by 3.38, from 1.2 to 4.05 years, so a city sizing its trainee cohort from certification hours provisions four trainee posts where fourteen are required.

Keywords: human–AI–robot task allocation, municipal automation governance, capability preservation, civic automation debt, workforce skill formation, algorithmic contestability, urban digital twin, constrained optimization

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1 INTRODUCTION

Urban automation has changed character. Traditional smart-city systems sensed, connected, and analyzed urban data. Agentic systems additionally plan, decide, negotiate, invoke tools, and act, while autonomous robots physically execute those decisions in public space. Choi and Yoon (2025) demonstrate AI-agent-based urban digital twins operating over live municipal data, and Shariatpour and Behzadfar (2026) describe the resulting shift toward "agentic cities" in which urban infrastructures sense, learn, predict, and act.

This creates a distributional question that allocation mechanisms do not yet address. A bridge inspection may be performed by a licensed inspector, a computer-vision agent, an autonomous robot, or a mixed team; a citizen service request may be handled by a public employee, an AI agent, or a human supervised by one; and sanitation, maintenance, transport, logistics, emergency response, and administration decompose in the same way. Each such decision gives work to some parties and withdraws it from others, and we know of no municipal allocation mechanism that records who bears the loss.

The policy problem is not whether AI and robots will replace municipal workers, but how a city should continuously decide which tasks are automated, augmented, reserved for humans, or performed by hybrid teams, and who performs the work that remains. We treat the second part, who performs the remaining work, as part of the allocation decision rather than its by-product. Automation exposure is distributed unevenly (Acemoglu and Restrepo, 2020; Eloundou et al., 2024), and nothing ensures that the developmental work that turns a novice into an expert is distributed evenly across occupational classes, genders, contract statuses, or districts. When a municipality automates entry-level inspection, triage, or casework, it removes the practice through which junior staff become competent and narrows the entry route for later cohorts. Because these decisions are made one locally defensible task at a time, a productivity objective evaluated task by task does not register the loss.

Partial automation also dilutes the tasks that remain: a junior practitioner working beside a lead on machine-assisted tasks accrues less practice per clock hour, so competence takes longer. In the bridge-inspection case of Section 6, machine assistance and shared supervision together extend formation from the 1.2 years that certification hours imply to 4.05 years, so a municipality that sizes its trainee cohort from certification hours provisions four trainee posts where fourteen are required (Figure 3).

1.1 Contributions

Most components adapt established constructions, as Section 2 records; the contribution lies in the decision variable that joins them and in what the joining makes computable.

- A city-scale allocation formalism whose decision variable is a *staffed mode*: a pairing of an execution mode with a roster of named practitioners at declared career stages. This makes distributional questions expressible in the model, since mode selection alone cannot say who receives the work.
- *Civic Automation Debt*, a penalty for five deferred liabilities: lost skill formation, eroded fallback, diluted accountability, vendor dependency, and unsupported displacement. Bainbridge (1983) stated the concern for individual operators; we aggregate it across a municipal workforce, separate it formally from the objective's operational flows, and give it a computable switching threshold.
- A *Human Capability Preservation Budget* in qualified-practice hours that accrues only to practitioners below independent competence, with a stated unit conversion from raw certification hours and an explicit feasibility condition.
- A *trainee-cohort requirement* and *dilution factor*, which turn the budget into the number of developing practitioners a domain must hold and show, for the worked domain, that machine assistance and shared supervision slow the formation of each practitioner (Section 6.6).
- A *Capability Access Constraint* governing who receives protected practice, so that a budget sized from the incumbent workforce reproduces its composition only by recorded decision, with the equality-law and data-protection analysis this requires.
- A *Policy Digital Twin* with a rule language, status lattice, and conflict order, and *Adaptive Oversight Zones* with specified transition predicates.
- A city-wide assignment program, an online Lagrangian approximation with a bounded between-rebalance violation, and a bridge-inspection allocation computed end to end by exact arithmetic, with a sensitivity sweep and two parameter shocks.

- An evaluation protocol specified in full, with ten pre-specified criteria: eight hypotheses with refutation criteria and two benchmarks calibrated against the worked example. It has not been executed, and no claim here rests on it.

Sections 2 to 7 give, in order, the related work and research questions, the framework's layers, the program and formal results, governance and legality, the worked allocation, and the discussion, protocol, and limitations.

2 RELATED WORK AND RESEARCH GAP

This section states our positioning; Appendix S7 gives the survey work by work.

2.1 Task Allocation and Workforce Scheduling

Capability-based allocation matches task requirements to agent capabilities, so that allocation follows task–agent fit rather than automation for its own sake (Ranz et al., 2017; Dorri et al., 2018). Later work added ergonomics (El Makrini et al., 2019; Merlo et al., 2023), human preferences (Noormohammadi-Asl et al., 2025), and measured trust (Ali et al., 2022), with reviews covering robot fleets (Meseguer Valenzuela and Blanes Noguera, 2025) and multi-agent reinforcement learning (Hady et al., 2025); the closest city-scale treatment allocates sensing tasks across a mobile crowd as a graph-structured assignment (Zhao et al., 2023). All of it optimizes an operational team's performance. None asks whether repeated local decisions erode the experience that sustains future expertise, or whose expertise erodes first.

The capability budget of Section 3.5 belongs to the long operations-research tradition of constrained training-hour allocation rather than departing from it: nurse rostering carries skill mix, supervision ratios, and qualification maintenance as hard constraints (Burke et al., 2004), and personnel scheduling includes cross-training variants (Van den Bergh et al., 2013). Two differences matter. First, traditional rostering assumes the task exists and asks who covers it, whereas our decision is prior (determining whether a human performs it at all, and thus whether developmental content exists to allocate). Second, rostering uses qualification requirements to keep today's coverage feasible, whereas our budget is sized from attrition to form the practitioners who will replace those leaving, and the objective also carries the deferred liabilities of Civic Automation Debt.

2.2 Automation, Skill Erosion, and Capability Formation

Automating the routine components of an operational task degrades human intervention capabilities (Bainbridge, 1983), a mechanism parameterized in our fallback erosion penalty D^{fall}. The out-of-the-loop problem (Endsley and Kiris, 1995), automation failure taxonomies (Parasuraman and Riley, 1997), and formal levels-of-automation scales (Sheridan and Verplank, 1978; Parasuraman et al., 2000) inform our adaptive oversight tiering (Zones Z1–Z4), while Shneiderman (2020) argues that automation and human control operate along independent dimensions. Zuboff's (1988) distinction between automating and informating work separates tasks that displace practitioners from those that build competence. Skill-decay meta-analyses (Arthur et al., 1998; Tatel and Ackerman, 2025) measure retention as a function of the practice interval, and would calibrate D^{skill} if it were extended with a retention term (Section 7.6).

CivicWorkOS therefore does not claim to have discovered skill erosion; Rashidi (2026) argues that AI is the first technology able to automate the entry-level cognitive work through which professional judgment is formed. That literature works at the scale of the individual operator, where Bainbridge (1983) showed that the ability to intervene decays separately from routine competence; our question, whether a city's expertise pipeline is drawn down across thousands of independent decisions, is municipal, and D^{fall} carries that separation into it. The debt framing has a source too: technical debt as the deferred cost of an expedient choice (Cunningham, 1992; Sculley et al., 2015), here an institutional liability deferred to future municipal administrations, absent from the operational ledger that authorized it.

2.3 Fair Division and Algorithmic Due Process

Protected practice is a scarce, unequally distributed good, placing the Capability Access Constraint inside the fair-division literature (Bouveret et al., 2016; Caragiannis et al., 2019). We use a share constraint against a policy-set target rather than an envy-based criterion because a panel must defend a distribution

against a statutory equality duty, not agents' valuations, and a Nash-welfare objective would need worker preferences that no city records and workers would report strategically. The cost is that our constraint offers no envy guarantee.

A framework that weighs civic costs must also show its work, or the weighting is unfalsifiable in the way opaque public-sector scoring is (Eubanks, 2018). Technological due process requires notice, a record, and a route to challenge (Citron, 2008; Citron and Pasquale, 2014), and a right to human intervention is empty unless a human can alter the outcome (Almada, 2019; Vaccaro et al., 2019). Of the contestability features Alfrink et al. (2023) synthesize, Section 5.2 implements the evidentiary record, the intervention request, and challenge to the rules themselves; Article 22 GDPR (European Parliament and Council of the European Union, 2016) sets the EU floor.

2.4 Algorithmic Management, Labor, and Who Bears the Cost

AI is becoming a collaborator (Senjic et al., 2026) and a manager: algorithmic management already allocates work, issues instructions, and monitors performance (Milanez et al., 2025), with gains in consistency set against autonomy, privacy, accountability, and discrimination concerns (OECD, 2026) that persist even where automation performs well (Wischnewski et al., 2023). Municipalities differ because public services are constituted by the discretionary judgment of street-level bureaucrats (Lipsky, 2010), so removing discretion changes what the service is. Ethical frameworks (Callari et al., 2024), the EU AI Act's oversight requirement (European Parliament and Council of the European Union, 2024), and the ISO 10218 series (International Organization for Standardization, 2025a,b) specify principles or physical constraints, not a mechanism that turns them into real-time decisions about who performs a task.

On employment, the task-based framework (Autor et al., 2003) underpins the argument that displacement is overstated when complementarities are ignored (Autor, 2015), and occupation-level exposure estimates (Frey and Osborne, 2017) fall roughly fivefold when re-measured at task level (Arntz et al., 2016); only task redesign is actionable by an allocation mechanism. Language models widen exposure (Eloundou et al., 2024), robot adoption lowers employment and wages for lower-skilled workers (Acemoglu and Restrepo, 2020), and platform workers bear their exposure outside the protections municipal employment presumes (Wood et al., 2019; De Stefano, 2016). Residents face a different exposure: automated public-service decisions concentrate their errors on the poorest (Eubanks, 2018). The ILO just-transition guidelines specify what an institutional response must contain (International Labour Organization, 2015), alongside OECD policy (OECD, 2026), employer reskilling forecasts (World Economic Forum, 2025), and Arab-region projections that Section 5.4 takes up (International Labour Organization and United Nations Economic and Social Commission for Western Asia, 2026).

We use *capability* in the sense of Sen (1999) and Nussbaum (2011), the real freedom to do and to be that a developmental task confers and an automated one withholds, and the budget takes the institutional form Ostrom (1990) identifies for renewable but depletable shared resources: monitored, bounded, locally governed appropriation rules. We borrow their concepts, not a decision procedure, which none of them was meant to supply.

2.5 Agentic-City Governance and Urban Digital Twins

Public-space robot policy concentrates on safety and privacy over wider wellbeing (Mintrom et al., 2026); co-creation and transparent civic platforms raise measured trust in municipal AI (Phothong et al., 2026; Nechesov and Ruponen, 2024), and the workforce's own digital readiness conditions reform (Aldhi et al., 2025). City digital twins already assist policy decisions (Temple et al., 2024), and Diaz-Sarachaga (2025) supplies criteria for whether a deployed twin is well governed, the closest antecedent to our Policy Digital Twin. We ask the converse: whether governance can be represented inside the twin as an executable constraint, carrying the social dimensions twins usually omit (Ben-Daya et al., 2026; Qanazi et al., 2025).

The resilience reserve of Section 3.7 draws on a literature that uses the term inconsistently (Mentges et al., 2023); response networks recover faster when post-failure reassignment is planned in advance (Lian et al., 2025), which is what the reserve plans.

2.6 Research Gap and Research Questions

Table 1 positions CivicWorkOS against the closest published frameworks on the dimensions that matter for a distributional claim.

Table 1. Positioning of CivicWorkOS against the closest published allocation and governance frameworks. Columns: task–agent fit; multi-objective objective; deferred cost entered as a term of the objective; mechanism decides who receives the work; policy enters as an executable constraint; contestability procedure specified; primary scale of application. ✓ present and specified; ▲ partially present or asserted without mechanism; ✗ absent.

Framework	Fit	Multi-obj.	Debt term	Who works	Policy	Contest.	Scale
Ranz et al. (2017)	✓	✗	✗	✗	✗	✗	Workcell
El Makrini et al. (2019)	✓	▲	✗	✗	✗	✗	Workcell
Merlo et al. (2023)	✓	✓	✗	✗	✗	✗	Workcell
Ali et al. (2022)	✓	▲	✗	✗	✗	✗	Team
Zhao et al. (2023)	✓	✓	✗	✗	✗	✗	City sensing
Nurse rostering (Burke et al., 2004)	✓	✓	▲	✓	▲	✗	Department
Diaz-Sarachaga (2025)	✗	✗	✗	✗	▲	▲	City
Alfrink et al. (2023)	✗	✗	✗	✗	▲	✓	System
CivicWorkOS	✓	✓	✓	✓	✓	✓	City

Five gaps follow. They are gaps of integration more than of method: each construction we use has a precedent above, and what is missing is their joining at municipal scale. First, allocation models and policy frameworks exist separately, and we know of no city-scale mechanism that makes policy an active constraint of allocation. Second, the single-operator mechanism of skill erosion has not been aggregated to a municipal workforce and entered into allocation.

Third, and most consequentially, rostering decides who works once the work is assigned to humans, but no mechanism we know of that makes the prior human-versus-machine decision also asks who receives the developmental work that decision preserves. We expect a preservation rule calibrated on the current workforce to reproduce its composition (gender balance, credentialing routes, neighborhoods, and the ratio of permanent to contracted staff), so that a rule meant to protect expertise would also keep out those not already represented.

Fourth, allocation needs a measure of whether enough human capability remains to recover essential services when AI, platforms, or robots fail. Fifth, automation rules are static compliance checks, whereas the acceptable degree of automation varies with sector, risk, labor conditions, and observed performance. These gaps motivate seven research questions: the first gap motivates RQ1 and RQ5, the second RQ2 and RQ3, the third RQ4, the fourth RQ6, and the fifth RQ7. Section 7.2 states which are answered, and how.

RQ1 How can urban tasks be allocated among humans, AI agents, robots, and hybrid teams to satisfy both operational and civic objectives?

RQ2 Can Civic Automation Debt represent long-term capability losses without double-counting immediate outcomes?

RQ3 Can a capability budget sustain expertise pipelines without fixed quotas, and at what cost?

RQ4 Who receives protected developmental work, and can an access constraint prevent preservation from reproducing the incumbent workforce?

RQ5 How can governance requirements become machine-checkable constraints with an operable contestability procedure?

RQ6 Can a resilience reserve preserve critical services under automation failure?

RQ7 How should oversight zones respond to performance, incidents, workforce capability, and feedback?

3 THE CIVICWORKOS FRAMEWORK

3.1 Problem Statement, Staffed Modes, and Civic Automation Debt

Appendix S9.5 collects the notation. Consider a city containing urban tasks $\mathcal{T} = \{T_1, \ldots, T_n\}$ and a heterogeneous workforce of humans (H), software AI agents (A), physical robots (R), and their hybrid combinations, giving the execution-mode set $\mathcal{M}$ defined in Equation (1):

$$\mathcal{M} = \{H,\ A,\ R,\ H{+}A,\ H{+}R,\ A{+}R,\ H{+}A{+}R\}. \tag{1}$$

Each task belongs to one critical service $\sigma(i) \in \mathcal{S}$ and one capability domain $k(i) \in \mathcal{K}$, both taken from the municipal service catalog.

Practitioners, Career Stage, and Staffed Modes

Selecting a mode alone cannot express who receives the work, so we make the human staffing explicit. Let $\mathcal{W}_k$ be the practitioners on the municipal payroll qualified in domain k (Section 7.5 discusses those outside it), each carrying an accumulated qualified-practice total $c_w(t)$ and facing a competence threshold h_k, which gives the *career stage* of Equation (2):

$$\mathrm{stage}_k(w,t) = \begin{cases} \textit{developing}, & c_w(t) < h_k, \\ \textit{competent}, & c_w(t) \geq h_k. \end{cases} \tag{2}$$

Only competent practitioners may hold roles carrying statutory sign-off. $N_k(t)$ counts competent practitioners in domain k and $N_k^{dev}(t)$ the developing ones.

A mode m specifies $n_m \geq 0$ human slots. A *roster* a assigns those slots to distinct available practitioners with a work split $\omega_{a,w} \geq 0$, $\sum_{w \in a} \omega_{a,w} = 1$; modes with no human slot admit only the null roster. The pair (m, a) is a ***staffed mode***, and the decision variable is defined in Equation (3):

$$x_{i,m,a} \in \{0,1\}, \qquad \sum_{m \in \mathcal{M}} \sum_{a \in \mathcal{A}_{i,m}} x_{i,m,a} = 1 \quad \forall i, \tag{3}$$

where $\mathcal{A}_{i,m}$ is the set of feasible rosters. The *developmental eligibility* of a roster, Equation (4),

$$\psi_a(t) = \sum_{w \in a} \omega_{a,w} \mathbb{1}\Big[\mathrm{stage}_{k(i)}(w,t) = \textit{developing}\Big] \in [0,1], \tag{4}$$

is the share of the human work performed by practitioners still being formed, and the *group share* of Equation (5),

$$\pi_{a,g} = \sum_{w \in a} \omega_{a,w} \mathbb{1}[\,w \in g\,], \qquad \sum_{g \in \mathcal{G}} \pi_{a,g} = 1, \tag{5}$$

is well defined for a hybrid team drawn from several groups and makes the protected-practice shares of Section 3.6 sum to one by construction. Worker groups $\mathcal{G}$ are a coarsening of the protected attributes declared by the panel of Section 5.1 subject to a minimum cell size n^{min} (30 practitioners per cell per domain by default), not their Cartesian product; cells below n^{min} are reported descriptively and carry no binding constraint (Appendix S8.1).

Civic Automation Debt

An allocation chosen on time, cost, accuracy, or safety generates externalities invisible at the task level: repeated automation removes entry-level learning, concentrates knowledge in proprietary systems, erodes fallback capacity, shifts accountability, and imposes transition costs on identifiable workers. We define *Civic Automation Debt* (CAD) as a per-task penalty for these five deferred liabilities, each component a marginal proxy for the change a staffed mode causes in a municipal stock. For task T_i under candidate

staffed mode (m, a) it is Equation (6):

$$CAD_{i,m,a} = \alpha\, D^{skill}_{i,m,a} + \beta\, D^{fall}_{i,m} + \gamma\, D^{acct}_{i,m} + \delta\, D^{dep}_{i,m} + \epsilon\, D^{trans}_{i,m,a}, \tag{6}$$

where D^{skill} is skill-formation loss, D^{fall} loss of human fallback capability, D^{acct} accountability dilution, D^{dep} technological dependency, and D^{trans} labor-transition burden, with weights summing to one (Section 4.3).

If the five CAD penalty components restated flow quantities already present in the objective, the model would double-count. We separate them by measurement domain. The other terms of the objective, Equation (27) of Section 4.3, represent *operational flows on task T_i's execution window*, whereas the CAD components act as *marginal proxies for changes in municipal stocks* whose accumulation Section 7.1 models. Equity appears twice but as two distinct objects: Eq^{srv} is the equity of *service to residents* in this window, while D^{trans} evaluates the transition burden on *workers* over the settlement horizon. Developmental content reaches the decision through four distinct channels, which Appendix S8.7 separates and Section 4.7 weighs.

Each D term is normalized to $[0, 1]$ and estimable from signals a city already tracks (Appendix S8.2). Skill-formation loss is the one that carries the framework, Equation (7):

$$D^{skill}_{i,m,a} = 1 - \phi_m\, \psi_a, \tag{7}$$

where ϕ_m is the share of the task's developmental content a human experiences under mode m (Section 3.5). Both factors are needed: a task performed entirely by machine forms nobody, and neither does one performed by an already-competent practitioner alone.

3.2 Layer 1: Urban Task Digital Twin

Every service is decomposed into tasks carrying a requirement vector and a duration, Equation (8):

$$T_i = \{cog_i, phy_i, emp_i, risk_i, auth_i, priv_i, urg_i, learn_i, crit_i;\ d_i\}, \tag{8}$$

covering cognitive, physical, and social demand, safety risk, legal-authority requirement, privacy sensitivity, urgency, human learning value, and criticality, each normalized to $[0, 1]$, and the execution time d_i in hours (Appendix S8.3). The task's *developmental content*, Equation (9),

$$\ell_i = learn_i \cdot d_i \quad [\text{qualified-practice hours}], \tag{9}$$

is the unit in which the capability budget is denominated. Its factor $learn_i$, like $crit_i$, is an institutional determination rather than physical telemetry, which is where the twin departs from the ISO 23247 reference architecture for digital twins of physical systems (Shao et al., 2023).

3.3 Layer 2: Civic Capability Ledger

The Civic Capability Ledger records, for each domain k, the state of Equation (10):

$$L_k(t) = \big\{H_k(t), A_k(t), R_k(t), F_k(t), E_k(t),\ \{E_{k,g}(t)\}_{g\in\mathcal{G}},\ \{c_w(t)\}_{w\in\mathcal{W}_k}\big\}, \tag{10}$$

comprising available human, AI, and robotic capability, fallback capacity F_k, the expertise pipeline E_k and its decomposition by worker group, without which the access constraint cannot be evaluated, and the per-practitioner totals that (2) reads. Fallback capacity is defined in Equation (11):

$$F_k(t) = \sum_{w\in\mathcal{W}_k} \zeta_w(t)\, \mathbb{1}[\text{stage}_k(w, t) = \textit{competent}]\,, \tag{11}$$

the qualified hours deployable under surge within a service's recovery window, with $\zeta_w(t)$ practitioner w's surge-deployable hours net of committed work. Because it counts only competent practitioners, F_k falls whenever they leave without being replaced from the pipeline.

3.4 Layer 3: Policy Digital Twin

The Policy Digital Twin converts laws, rules, collective agreements, and local priorities into machine-checkable constraints. A policy rule is the tuple of Equation (12):

$$\rho = \langle \text{scope},\ \text{guard},\ \text{status},\ \text{authority},\ \text{validity},\ \text{provenance} \rangle, \tag{12}$$

where the scope is a conjunction of catalog predicates, the guard a decidable predicate over the task vector, roster, and ledger state, the authority ranks the source from statute down to operational guidance (it governs amendment, not evaluation; Appendix S8.6), the validity is a time interval, and the provenance names the instrument and the panel decision that encoded it. Statuses form the totally ordered lattice of Equation (13),

$$\textit{allow} \prec \textit{allow-with-oversight} \prec \textit{restrict} \prec \textit{prohibit}. \tag{13}$$

Policy is evaluated in two stages, so that rules about modes can be checked before any roster exists: a mode-level status

$$\mathcal{P}_{mode}(T_i, m, t) \ = \bigsqcup_{\rho \in \text{fires}_{mode}(T_i,m,t)} \text{status}(\rho), \tag{14}$$

evaluated before roster assembly, and a roster-level status

$$\mathcal{G}_{roster}(T_i, m, a, t) \ = \bigsqcup_{\rho \in \text{fires}_{roster}(T_i,m,a,t)} \text{status}(\rho), \tag{15}$$

evaluated for each candidate roster in $\mathcal{A}_{i,m}$. The joint status for staffed mode (m, a) is then the lattice join of Equation (16):

$$\mathcal{P}(T_i, m, a, t) \ = \ \mathcal{P}_{mode}(T_i, m, t) \sqcup \mathcal{G}_{roster}(T_i, m, a, t) \ \in \ \{\textit{allow}, \textit{allow-with-oversight}, \textit{restrict}, \textit{prohibit}\}. \tag{16}$$

The most restrictive applicable rule therefore takes precedence, which is the appropriate default for a public body. Because the join ignores the source of a rule, no rule can relax another, whatever its authority; relaxing a rule means amending or retiring it through the panel. The two intermediate statuses are executable: *allow-with-oversight* binds a named reviewer to the staffed mode and adds the reviewer's time and retained judgment to its terms, while *restrict* narrows $\mathcal{A}_{i,m}$ to rosters meeting a minimum human share or stage requirement, which is how a supervision ratio enters the model. Appendix S8.4 gives the conflict order, the consistency check, and a worked encoding of Article 14 of the EU AI Act (European Parliament and Council of the European Union, 2024); Section 5.2 makes the twin's decisions contestable.

3.5 Layer 4: Human Capability Preservation Budget

The Human Capability Preservation Budget (HCPB) reserves a minimum volume of developmental human practice, denominated throughout in *qualified-practice hours per budget period* ΔT, taken as one year, so that both sides of the constraint are in one auditable unit. The *human developmental share* of mode m is given in Equation (17):

$$\phi_m \in [0, 1], \qquad \phi_H = 1, \quad \phi_A = \phi_R = \phi_{A+R} = 0, \tag{17}$$

with hybrid values set per domain from an assessment of how much of the task's judgment a human still exercises. The HCPB constraint is Equation (18):

$$\Lambda_k(\Delta T) \ = \sum_{i \in \mathcal{T}_k(\Delta T)} \ \sum_{m \in \mathcal{M}} \ \sum_{a \in \mathcal{A}_{i,m}} x_{i,m,a}\, \ell_i\, \phi_m\, \psi_a \ \geq \ B_k(t), \tag{18}$$

both sides in hours delivered to practitioners below competence. The factor ψ_a restricts credit to practitioners still being formed: without it a domain that gives every task to its incumbents satisfies a budget sized from

novice formation while forming nobody. The budget is estimated in Equation (19):

$$B_k(t) = \max\Big(B_k^{min},\ \eta_k\, N_k(t)\, r_k(t)\, h_k\Big), \tag{19}$$

where $r_k(t)$ is the fraction of competent practitioners lost per period, h_k the qualified-practice hours that bring one developing practitioner to competence, η_k a policy-set replacement factor ($\eta_k > 1$ builds margin, $\eta_k < 1$ deliberately lets a domain shrink), and B_k^{min} an hours floor. Certification schemes count *raw clock hours* h_k^{raw} of logged practice. We read these as hours in which the candidate performs all of the human work of the task without machine assistance, a competent supervisor signing off; such hours accrue the mean learning value $\overline{learn}_k$ per clock hour, so Equation (20) converts between them:

$$h_k \ = \ \overline{learn}_k \cdot h_k^{raw}. \tag{20}$$

A practitioner who works on machine-assisted tasks, or shares the human work with a lead, accrues less than $\overline{learn}_k$ per clock hour, which is why realized formation takes longer than certification hours imply (Section 6.6). The budget is not a jobs quota: it constrains a volume of practice rather than a number of posts, so a task may go to a developing human supported by an AI agent even where AI could perform it alone. In Ostrom's terms (1990) it is an appropriation rule over a renewable but depletable shared resource, bounded, monitored, and revised through the panel of Section 5.1.

3.6 Layer 4b: Capability Access and Just Transition

Equation (18) decides *how much* developmental work is protected but not *who* performs it, and left alone its answer depends on recruitment: if developing practitioners are recruited through the pipelines that produced the incumbents, a composition-blind budget reproduces the incumbent composition for a full replacement cycle under the name of preserving capability. We therefore separate the budget's *size*, derived from attrition, from its *distribution*, which must be set deliberately. The protected practice share of group g is defined in Equation (21):

$$\Pi_{k,g} = \frac{\sum_{i\in\mathcal{T}_k(\Delta T)} \sum_m \sum_a x_{i,m,a}\, \ell_i\, \phi_m\, \psi_a\, \pi_{a,g}^{dev}}{\sum_{i\in\mathcal{T}_k(\Delta T)} \sum_m \sum_a x_{i,m,a}\, \ell_i\, \phi_m\, \psi_a} \tag{21}$$

where $\pi_{a,g}^{dev}$ is the group share (5) restricted to the roster's developing members, so that $\sum_g \Pi_{k,g} = 1$. The *Capability Access Constraint* is Equation (22):

$$\Pi_{k,g} \ \geq \ \theta_{k,g} - \varepsilon_k \qquad \forall\, g \in \mathcal{G}, \tag{22}$$

with target composition $\theta_{k,g}$ and tolerance ε_k. The critical design choice is that $\theta_{k,g}$ is *not* the incumbent composition. It is set by the panel of Section 5.1 from the qualified local labor pool, a statutory equality duty, or an explicit equity target, and published with the weight vector; setting it to the incumbent share remains permitted, but is then a recorded decision to freeze composition rather than an unnoticed property of the arithmetic. Feasibility requires $\sum_g(\theta_{k,g} - \varepsilon_k) \leq 1$, which the twin checks on publication. Section 5.3 addresses whether such a target may lawfully be set.

The second distributional constraint concerns those who lose work. With $\Delta_g(\mathbf{x}) = \big[H_g^{base} - \sum_{i\in\mathcal{T}(\Delta T)} \sum_m \sum_a x_{i,m,a}\, d_i \sum_{w\in a\cap\mathcal{W}_g} \omega_{a,w}\big] / H_g^{base}$ the net fractional reduction in the allocated hours of payroll group g relative to its baseline H_g^{base}, the *Just Transition Constraint* is Equation (23):

$$\Delta_g(\mathbf{x}) \leq \tau_g \qquad \forall\, g \in \mathcal{G}, \tag{23}$$

a ceiling on the pace of displacement for any single group, following the ILO guidelines (International Labour Organization, 2015). The ceiling stops the framework from concentrating displacement on the group with least bargaining power, but only among payroll groups, since $\mathbf{x}$ assigns no hours to anyone else. Reskilling coverage is a municipal budget fact rather than a function of $\mathbf{x}$, so it is enforced in the

governance layer as a gating predicate rather than in the program (Appendix S8.5). Together, (22) and (23) do not resolve who wins and who loses, but make the answer explicit, versioned, auditable, and contestable.

3.7 Layer 5: Reversibility and Resilience Reserve

For each critical service s the city maintains a minimum recoverable capacity that does not depend on a single automation stack. Total service capacity comprises the automation-dependent components $\mathcal{C}_s^{aut}$ (an AI agent pool, a robot fleet class, a vendor platform) and the human contribution of the domains serving s, Equation (24):

$$C_s(t) \;=\; \sum_{c \in \mathcal{C}_s^{aut}} C_{s,c}(t) \;+\; \sum_{k \in \mathcal{K}_s} \xi_{s,k}\, F_k(t), \tag{24}$$

with $\xi_{s,k}$ the share of domain k's surge-deployable hours that s may draw. Surviving capacity after losing the n_s largest automation-dependent components $\mathcal{C}_s^{(n_s)}$ is Equation (25):

$$Res_s^{(n_s)}(t) \;=\; C_s(t) \;-\; \sum_{c \in \mathcal{C}_s^{(n_s)}(t)} C_{s,c}(t), \tag{25}$$

so when the components lost are all of the service's automation, as when one AI service or fleet carries it, what survives is the human capacity in the second term of Equation (24); otherwise the remaining automation components also count. The *Reversibility and Resilience Reserve* constraint is Equation (26):

$$Res_s^{(n_s)}(t) \;\geq\; R_s^{min}, \qquad R_s^{min} = \kappa_s\, D_s^{peak}, \tag{26}$$

with contingency order $n_s \in \{1, 2\}$, peak demand D_s^{peak}, and $\kappa_s \in (0, 1]$ the share of that peak which must remain servable. This is the N-1 and N-2 criterion of reliability engineering (Jain et al., 2024; Ouyang, 2014) applied to a workforce of humans, AI agents, and robots. The allocation reaches $Res_s^{(n_s)}$ only through F_k: assignments commit practitioners' hours, lowering ζ_w, and move developing practitioners across the threshold of Equation (2); in program (28), $Res_s^{(n_s)}(\mathbf{x})$ is Equation (25) with F_k evaluated after the window's assignments. Starving a domain of human practice lowers $F_k(t)$ over the formation period, and breaches (26) once surviving capacity rests on the human term and that term falls below the threshold (Figure 1).

3.8 Layer 6: Policy-Aware Multi-Agent Allocation Market

Ten specialized agents score candidate staffed modes: five *capability agents* estimate what a mode can do, and five *civic agents* estimate its effects on workforce development, equity, privacy, and resilience, among them the Policy Guardian Agent, which executes Equation (16) and holds a veto (Figure 1). The market is a single-round call for estimates, a structured elicitation mechanism rather than a strategic game, and we make no equilibrium claim for it. Appendix S8.8 lists the agents, the terms each estimates, and the protocol, and Section 7.6 records the principal–agent exposure the design leaves open.

4 ARCHITECTURE, OPTIMIZATION PROGRAM, AND ALGORITHM

4.1 Reference Architecture

Figure 1 realizes the layers of Section 3 as an event-driven system in which a single feedback bus returns outcomes, incidents, and appeals from all three execution channels to both governance services. Robot execution sits behind fleet middleware enforcing ISO 10218-2 (International Organization for Standardization, 2025b), with the robots conforming to ISO 10218-1 (International Organization for Standardization, 2025a), so a policy-level error still meets the hardware-level safety interlocks those standards require.

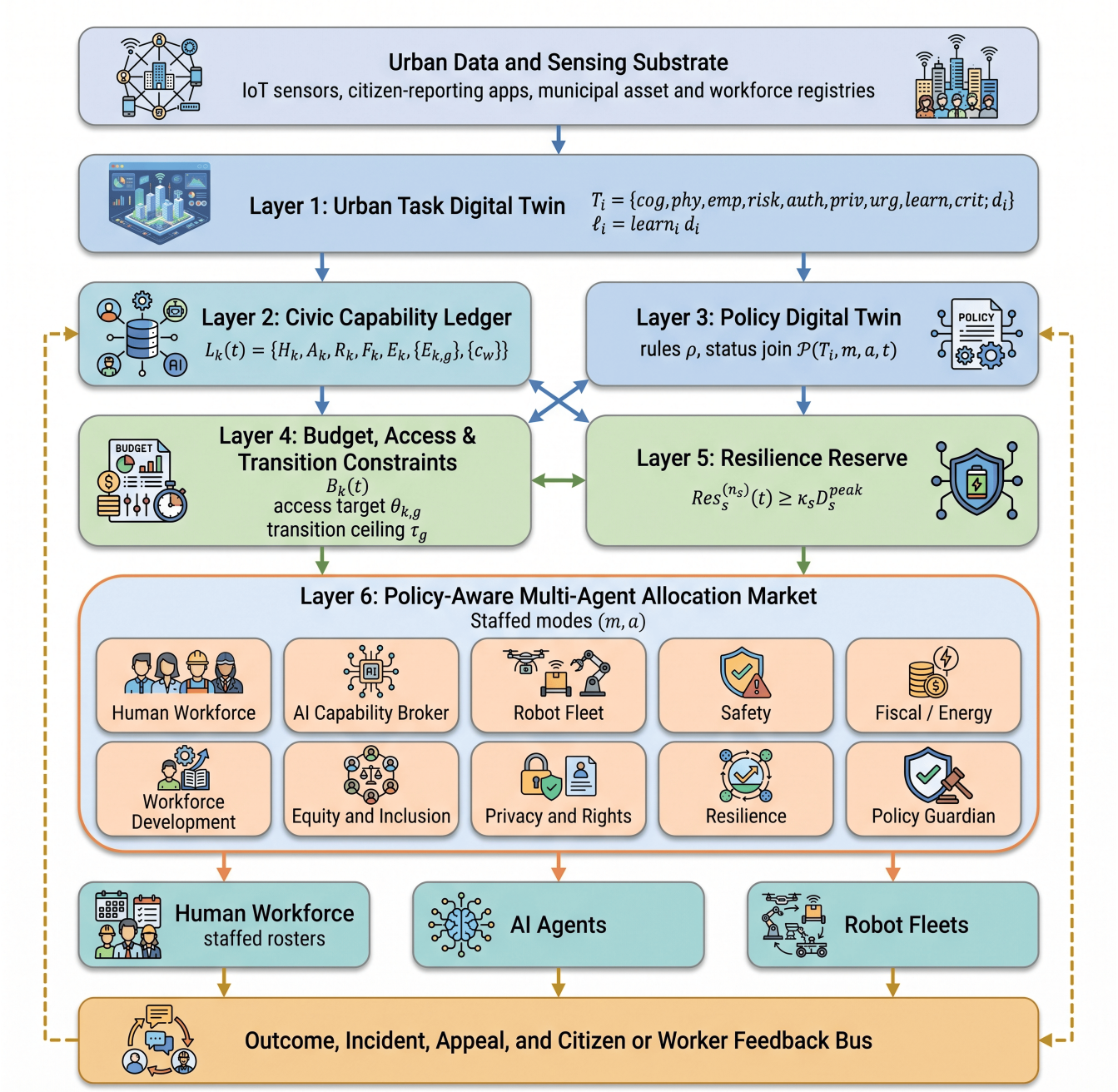


Figure 1. Reference architecture of CivicWorkOS, read top to bottom. Solid arrows are the forward allocation path from sensed urban data through the governance layers to execution; the box labeled Layer 4 covers Layers 4 and 4b of Section 3, and the ten agents of Section 3.8 are drawn inside the allocation market. The Ledger supplies fallback capacity F_k of Equation (11) to the reserve through Equation (24), and per-practitioner practice totals c_w to the career-stage test of Equation (2). The Policy Twin supplies the constraint layer with the replacement factor η_k, the access target $\theta_{k,g}$, and the roster guards that restrict $\mathcal{A}_{i,m}$ (these flows are not drawn as separate edges). Dashed arrows are the outcome-feedback path, which updates both governance services.

4.2 Runtime Allocation Workflow

Figure 2 shows the per-task control flow that Algorithm 1 formalizes. If policy filtering or roster pruning empties the candidate set, the task routes to Zone Z4 human-reserved execution and the event is counted, because its rate is the operational cost of the constraint system.

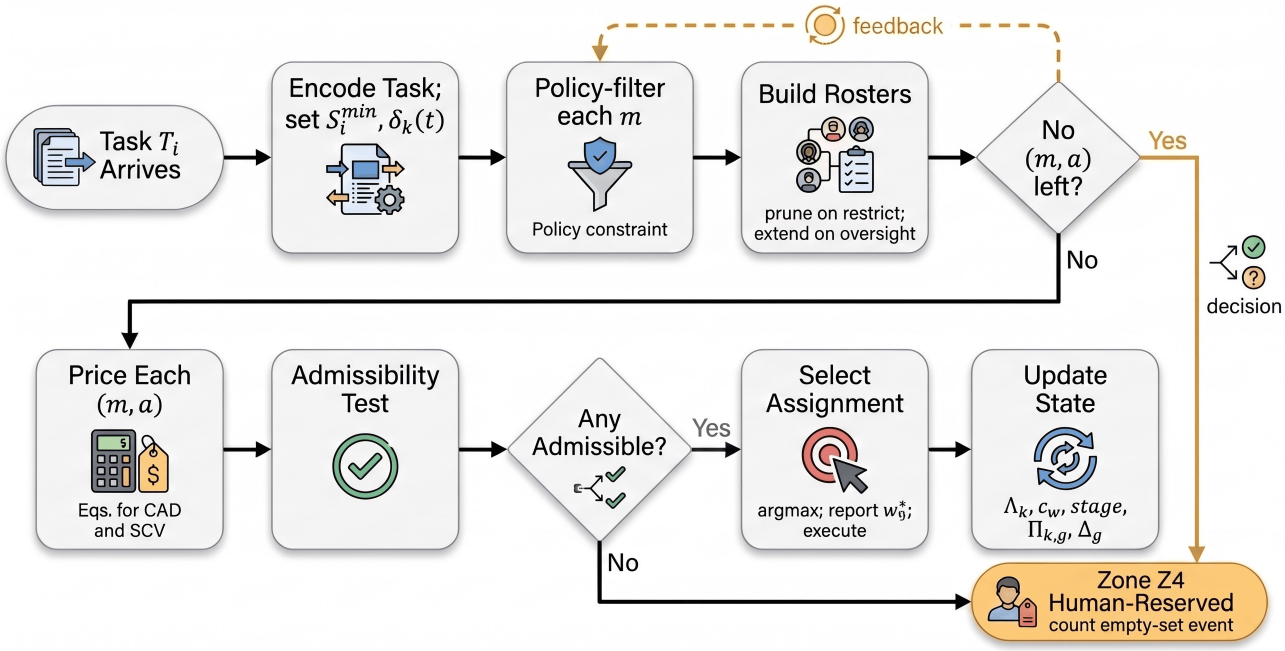


Figure 2. Runtime allocation workflow, matching Algorithm 1. The upper row encodes the task, sets the safety floor and the catch-up increment, policy-filters modes, and builds the roster set, pruning it where the twin returns *restrict* and extending it with a named reviewer where it returns *allow-with-oversight*. The lower row scores the surviving staffed modes (the box labeled "Price Each"), applies the admissibility test of Equation (31), and either falls back to Zone Z4 or selects and executes. The dashed *feedback* edge stands for Equation (34): outcomes return to the Policy Digital Twin after execution, which is what makes the zone-transition predicates (Section 4.8) evaluable.

4.3 Sustainable Civic Value (SCV)

For each task T_i and staffed mode (m, a), the *Sustainable Civic Value* is given by Equation (27):

$$\begin{aligned} SCV_{i,m,a} = {} & w_1 Q_{i,m,a} + w_2 S_{i,m,a} + w_3 P_{i,m,a} + w_4 Eq^{srv}_{i,m,a} + w_5 Tr_{i,m,a} \\ & - w_6 Cost_{i,m,a} - w_7 En_{i,m} - w_8 Pr_{i,m} - w_9 CAD_{i,m,a}, \end{aligned} \tag{27}$$

where Q is service quality, S safety, P productivity, Eq^{srv} the equity of service delivery across resident groups, Tr agent trust, $Cost$ operating cost, En energy use, Pr privacy risk, and CAD the debt of (6). The first eight terms are flows on the task's execution window and the ninth is a change in municipal stocks; six of the eight flow terms, and the debt term, depend on the roster as well as the mode.

A linear weighting trades flexibility for interpretability relative to Pareto-front methods (Osika et al., 2023; Zheng and Rasheed, 2026), at a cost to the panel's authority that Section 5.1 states. All nine terms are rescaled to $[0, 1]$: the flow terms against a trailing twelve-month window, and the debt components against externally fixed denominators. Against a rolling window a steadily decaying pipeline would be compared only with its own recent decline, and would keep scoring mid-range (Appendix S9.2).

Default Weights and Effective Weights

The illustrative weights are w_1=0.15 (quality), w_2=0.18 (safety), w_3=0.14 (productivity), w_4=0.10 (service equity), w_5=0.05 (trust), w_6=0.08 (cost), w_7=0.03 (energy), w_8=0.05 (privacy risk), and w_9=0.22 (CAD), with debt weights α=0.28, β=0.22, γ=0.18, δ=0.12, and ϵ=0.20. They are design defaults for the panel to revise, not fitted estimates; Table S3 assigns a default to every remaining parameter.

Table 2 reports the *effective* weight each construct carries once the weights are composed, because a weight vector can assert a priority the arithmetic withholds. The deferred block carries 0.22, more than any single flow term, but its largest component, skill formation at 0.062, ranks sixth, and the largest single

construct is safety at 0.180. Labor-transition burden carries only 0.044, deliberately: an objective weight can be outvoted, so transition is also carried by the hard constraint (23), and a city that judges displacement unacceptable lowers τ_g rather than raising $w_9\epsilon$. Proposition 3 shows whether the weight or a constraint is deciding.

Table 2. Effective weight of each construct in the CivicWorkOS default configuration, obtained by composing $w_9 = 0.22$ with the debt weights. The deferred block carries more weight than any single flow term; safety is the largest single construct. Column sums to 1.000.

Construct	**Enters via**	**Effective weight**
Safety	w_2	0.180
Service quality	w_1	0.150
Productivity	w_3	0.140
Service equity (residents)	w_4	0.100
Operating cost	w_6	0.080
Skill formation	$w_9\alpha$	0.062
Trust	w_5	0.050
Privacy risk	w_8	0.050
Fallback capacity	$w_9\beta$	0.048
Labor-transition burden (workers)	$w_9\epsilon$	0.044
Accountability	$w_9\gamma$	0.040
Energy	w_7	0.030
Vendor dependency	$w_9\delta$	0.026
Deferred block total	w_9	*0.220*

4.4 The City-Wide Allocation Program

The constraints of Sections 3.5 through 3.7 couple decisions across tasks, so over a rebalancing window CivicWorkOS solves program (28):

$$\max_{\mathbf{x},\boldsymbol{\varsigma}} \quad \sum_{i\in\mathcal{T}(\Delta T)} \sum_{m\in\mathcal{M}} \sum_{a\in\mathcal{A}_{i,m}} x_{i,m,a}\, SCV_{i,m,a} \;-\; M \sum_{k,g} \varsigma_{k,g} \tag{28a}$$

$$\text{s.t.} \quad \textstyle\sum_m \sum_a x_{i,m,a} = 1, \qquad \forall i, \tag{28b}$$

$$x_{i,m,a} = 0 \;\text{ if } \mathcal{P}(T_i, m, a, t) = \textit{prohibit}, \tag{28c}$$

$$x_{i,m,a} = 0 \;\text{ if } S_{i,m,a} < S_i^{min}, \tag{28d}$$

$$\textstyle\sum_{i\in\mathcal{T}_k} \sum_m \sum_a x_{i,m,a}\, \ell_i\, \phi_m\, \psi_a \geq B_k(t), \qquad \forall k, \tag{28e}$$

$$\textstyle\sum_{i\in\mathcal{T}_k} \sum_{m,a} x_{i,m,a}\, \ell_i\, \phi_m\, \psi_a\, [\pi_{a,g}^{dev} - (\theta_{k,g} - \varepsilon_k)] + \varsigma_{k,g} \geq 0, \quad \varsigma_{k,g} \geq 0, \qquad \forall k, g, \tag{28f}$$

$$\Delta_g(\mathbf{x}) \leq \tau_g, \qquad \forall g, \tag{28g}$$

$$Res_s^{(n_s)}(\mathbf{x}) \geq \kappa_s D_s^{peak}, \qquad \forall s, \tag{28h}$$

$$\textstyle\sum_i \sum_m \sum_a x_{i,m,a}\, u_{i,m,a,r} \leq U_r, \qquad \forall r, \tag{28i}$$

$$x_{i,m,a} \in \{0, 1\}. \tag{28j}$$

Here $S_i^{min} = risk_i\, crit_i$ is the task's safety floor, $u_{i,m,a,r}$ is the consumption of resource r (human hours by grade and stage, fleet hours, inference budget), and U_r its availability. The access constraint (28f) is linearized by multiplying the ratio of Equation (21) through by the domain's total practice volume $\Lambda_k(\mathbf{x})$, with the elastic slack $\varsigma_{k,g} \geq 0$ denominated in qualified-practice hours to avoid bilinearity. Penalized at a large M, an unattainable target yields a published shortfall rather than infeasibility and never compels an individual assignment, which Section 5.3 argues is legally necessary. The program is a binary

multidimensional assignment problem with side constraints, NP-hard in general, and is therefore solved periodically over a window; this article makes no scalability claim.

4.5 Online Allocation, Duals, and the Admissibility Test

Between rebalances, tasks must be decided on arrival, so CivicWorkOS approximates (28) online by Lagrangian relaxation of all five coupling constraints, with dual multipliers $\lambda_k, \mu_s, \nu_{k,g}, \varrho_g, \upsilon_r \geq 0$ for (28e), (28h), (28f), (28g), and (28i) from the most recent solution. The online score is Equation (29):

$$\begin{aligned}\widetilde{SCV}_{i,m,a} = SCV_{i,m,a} &+ \lambda_{k(i)}\,\ell_i\,\phi_m\,\psi_a + \mu_{\sigma(i)}\,\Delta^{res}_{\sigma(i)}(i,m,a) \\ &+ \sum_g \nu_{k(i),g}\,\ell_i\,\phi_m\,\psi_a\,\big[\pi^{dev}_{a,g} - (\theta_{k(i),g} - \varepsilon_{k(i)})\big] \\ &- \sum_g \varrho_g\,\delta^{disp}_g(i,m,a) - \sum_r \upsilon_r\,u_{i,m,a,r},\end{aligned} \tag{29}$$

where $\delta^{disp}_g(i,m,a)$ is the estimated increase in Δ_g that assigning task i to (m,a) causes. Because one task does not change a fleet's capacity, the resilience term is counterfactual and scaled by the task's share of service demand, Equation (30):

$$\Delta^{res}_s(i,m,a) \;=\; \frac{d_i}{D^{tot}_s(\Delta T)}\Big[\,Res^{(n_s)}_s\big(t \,\big|\, (m,a) \text{ generalized across } s\big) - Res^{(n_s)}_s(t)\,\Big], \tag{30}$$

where $D^{tot}_s(\Delta T)$ is the service's total task-hours in the window and the first term is the surviving capacity if every task of service s in the window were executed in (m,a). The duals come from the root LP relaxation, the standard construction for Lagrangian decomposition (Appendix S9.3). The integrality gap the solver reports measures how far the integer optimum lies from that relaxation; it does not bound the error of the per-arrival rule, which also depends on how stale the duals are (Section 7.3).

The budget and reserve constraints concern the state of a domain, not of a staffed mode, so evaluating them inside a loop over (m,a) removes every candidate or none, and a city below threshold would fail hardest in the state the framework exists to prevent. We therefore test the post-execution state and add a feasibility-restoration branch. Staffed mode (m,a) is *admissible* for task i at time t if it satisfies Equation (31):

$$\begin{aligned}&\mathcal{P}(T_i,m,a,t) \neq \textit{prohibit} \quad\text{and}\quad S_{i,m,a} \geq S^{min}_i, \quad\text{and}\\ &\Big[\,Res^{(n_s)}_s(t^+ \,|\, m,a) \geq R^{min}_s \;\text{ or }\; \big(Res^{(n_s)}_s(t) < R^{min}_s \;\wedge\; \Delta^{res}_s(i,m,a) \geq 0\big)\Big], \quad\text{and}\\ &\Big[\,\Lambda_k(t^+ \,|\, m,a) \geq \bar{B}_k(t) \;\text{ or }\; \ell_i\,\phi_m\,\psi_a \;\geq\; \delta_k(t)\Big], \quad\text{and}\\ &\Delta_g\big(t^+ \,|\, m,a\big) \leq \tau_g \;\;\forall g, \quad\text{and}\quad u_{i,m,a,r} \leq U_r - U^{used}_r(t) \;\;\forall r,\end{aligned} \tag{31}$$

with $s = \sigma(i)$, $k = k(i)$, where $\Lambda_k(t)$ is the practice accrued in domain k so far in the current budget period, $\bar{B}_k(t) = B_k(t) \cdot t/\Delta T$ the pro-rated budget trajectory, and the catch-up threshold $\delta_k(t)$ is defined in Equation (32):

$$\delta_k(t) \;=\; \frac{\max\big(0,\; B_k(t) - \Lambda_k(t)\big)}{n^{rem}_k(t)} \tag{32}$$

the per-task increment that closes the remaining gap over the $n^{rem}_k(t)$ tasks still expected in the period. Admitting any mode with $\phi_m\psi_a > 0$ would let a lagging domain stay behind indefinitely; requiring $\ell_i\phi_m\psi_a \geq \delta_k(t)$ makes each mode carry its share of the shortfall, so the test tightens as the gap grows. As Λ_k crosses $\bar{B}_k(t)$ the test can switch abruptly between automated execution and trainee reservation; a hysteresis band or a proportional–integral filter on $\delta_k(t)$ would smooth it, but neither is specified or

analyzed here. The allocation is then determined by Equation (33):

$$(m,a)_i^* = \arg\max_{(m,a)\in\mathcal{A}(i,t)} \widetilde{SCV}_{i,m,a}, \tag{33}$$

over the admissible set $\mathcal{A}(i,t)$, with the Zone Z4 fallback when $\mathcal{A}(i,t)=\emptyset$. After execution, outcomes are folded back into policy through Equation (34), the closed governance loop of Section 5.1:

$$Policy_{t+1} = f\big(Policy_t,\, Performance_t,\, Incidents_t,\, CAD_t,\, Distribution_t,\, Feedback_t\big). \tag{34}$$

Algorithm 1 states the per-task procedure, including the roster construction and the two intermediate policy statuses.

Algorithm 1 Online policy-aware allocation over staffed modes

Require: Task T_i; time t; ledger $L(t)$; policy twin $\mathcal{P}$; budgets $B(t)$; thresholds R^{min}; duals $\lambda,\mu,\nu,\varrho,\upsilon$ from the last solution of (28)
Ensure: Selected staffed mode $(m,a)_i^*$

Encode T_i via (8); $\ell_i \leftarrow learn_i d_i$; $s \leftarrow \sigma(i)$; $k \leftarrow k(i)$; $S_i^{min} \leftarrow risk_i\, crit_i$
$\mathcal{A} \leftarrow \emptyset$; $\delta_k \leftarrow$ (32)
for each $m \in \mathcal{M}$ **do**
  $z \leftarrow \mathcal{P}_{mode}(T_i, m, t)$ ▷ mode-level screening via (14)
  if $z =$ *prohibit* **then**
    continue
  end if
  $\mathcal{A}_{i,m} \leftarrow$ feasible rosters for m from available $w \in \mathcal{W}_k$, with ψ_a by (4), $\pi_{a,g}^{dev}$ by (5)
  remove each a with $\mathcal{P}(T_i,m,a,t) =$ *prohibit* by (16), evaluating $\mathcal{G}_{roster}$ (15) for every a
  if $z =$ *restrict* **then**
    keep only rosters meeting the restricting rules' human-share or stage requirement
  end if
  if $z =$ *allow-with-oversight* **then**
    append the named reviewer to each roster; add reviewer time to $Cost$; raise ϕ_m by the assessed retained-judgment share
  end if
  for each $a \in \mathcal{A}_{i,m}$ **do**
    Query the ten agents for $Q, S, P, Eq^{srv}, Tr, Cost, En, Pr$ and $D^{skill}, D^{fall}, D^{acct}, D^{dep}, D^{trans}$
    $CAD_{i,m,a} \leftarrow$ (6); $SCV_{i,m,a} \leftarrow$ (27); $\Delta_s^{res}(i,m,a) \leftarrow$ (30)
    if (m,a) satisfies (31) **then**
      $\widetilde{SCV}_{i,m,a} \leftarrow$ (29); $\mathcal{A} \leftarrow \mathcal{A} \cup \{(m,a)\}$
    end if
  end for
end for
if $\mathcal{A} = \emptyset$ **then**
  $(m,a)_i^* \leftarrow$ Zone Z4 human-reserved roster; record the empty-set event; update $L(t)$; **return**
end if
$(m,a)_i^* \leftarrow \arg\max_{(m,a)\in\mathcal{A}} \widetilde{SCV}_{i,m,a}$ ▷ Equation (33)
Report $w_9^\star(i)$ by Proposition 3 to the allocation record
Execute; observe outcomes; $\Lambda_k \leftarrow \Lambda_k + \ell_i \phi_{m^*} \psi_{a^*}$
For each $w \in a^*$: $c_w \leftarrow c_w + \ell_i \phi_{m^*} \omega_{a^*,w}$; re-evaluate $\mathrm{stage}_k(w,t)$ by (2)
Update $L(t{+}1)$, accumulated CAD, $\Pi_{k,g}$, Δ_g, U_r^{used}; emit to the feedback bus via (34)

Per task, Algorithm 1 performs at most seven mode-level policy queries, one roster-level query and one round of agent estimates per candidate roster, $\sum_m |\mathcal{A}_{i,m}|$ in all, so the online decision is linear in arrivals and the periodic program is the bottleneck. The rule is exact only when the duals are exact and the state has not drifted (Section 7.3).

4.6 Feasibility of the Capability Budget and the Intake Requirement

A capability budget the task stream cannot supply is not demanding but infeasible. Proofs of the four propositions are in Appendix S9.

PROPOSITION 1 (Feasibility of the capability budget). *Let* $\Phi_k(\Delta T) = \sum_{i \in \mathcal{T}_k(\Delta T)} \ell_i$ *be the domain's total developmental content in the period, and let*

$$\bar{\zeta}_k \;=\; \max\big\{\phi_m \psi_a \;:\; (m,a) \text{ passes the policy and safety conditions of } (31) \text{ for some } i \in \mathcal{T}_k\big\}$$

be the largest developmental credit any such staffed mode delivers. Then Equation (18) *is satisfiable only if the condition in Equation* (35) *holds:*

$$B_k(t) \;\le\; \Phi_k(\Delta T)\,\bar{\zeta}_k. \tag{35}$$

Where the estimator (19) *is attained by its second argument, this is equivalent to a ceiling on attrition given by Equation* (36)*:*

$$r_k(t) \;\le\; r_k^{crit} \;=\; \frac{\Phi_k(\Delta T)\,\bar{\zeta}_k}{\eta_k\, N_k(t)\, h_k}, \tag{36}$$

and symmetrically to ceilings on η_k *and* h_k *and a floor on the task volume.*

Condition (35) is more restrictive than it looks: machine assistance and supervision ratios both reduce $\bar{\zeta}_k$, and safety floors can remove the high-ϕ_m candidates, as they remove H and $H{+}A$ in Section 6.1. When it fails, the framework raises an infeasibility flag, clamps B_k to $\Phi_k\bar{\zeta}_k$, publishes the shortfall as a capability deficit, and reports the four restoring actions: relax the supervision ratio where risk permits, procure less autonomous tooling, expand the supervised task stream, or lower η_k as a recorded decision to let the domain shrink. The next proposition turns the budget into the number of developing practitioners a domain must hold.

PROPOSITION 2 (Required trainee cohort). *Let* Θ_k *be the clock hours a developing practitioner works on domain tasks per budget period, and let* $\overline{learn}_k$*,* $\bar{\phi}_k$*, and* $\bar{\omega}_k$ *be the duration-weighted means, over those hours, of the task learning value* $learn_i$*, the human developmental share* ϕ_m *of the selected mode, and the practitioner's work share* $\omega_{a,w}$ *in the selected roster. Assume the duration-weighted mean of the product* $learn_i\,\phi_m\,\omega_{a,w}$ *equals* $\overline{learn}_k\bar{\phi}_k\bar{\omega}_k$*, as it does when the three are mutually independent across the assigned stream (pairwise zero covariance is not sufficient). Then the expected time for one developing practitioner to reach competence is given by Equation* (37)*:*

$$\tau_k \;=\; \frac{h_k^{raw}}{\Theta_k} \cdot \frac{1}{\bar{\phi}_k\,\bar{\omega}_k}, \tag{37}$$

and in steady state the domain must hold a standing cohort of

$$\hat{n}_k \;=\; \eta_k\, N_k(t)\, r_k(t)\, \tau_k \tag{38}$$

developing practitioners for (18) *to be met, a stock fed by* $\eta_k N_k r_k$ *completions per period.*

Equation (37) factorizes formation time into the time h_k^{raw}/Θ_k that certification hours imply and a *dilution factor* $1/(\bar{\phi}_k\bar{\omega}_k)$, with $1/\bar{\phi}_k$ due to machine assistance and $1/\bar{\omega}_k$ to sharing the human work with a lead. If supervisors assign high-learning tasks to rosters that give the trainee more of the human work, formation is faster than Equation (37) gives; if automated modes absorb the highest-learning tasks, it is

slower. Section 6.6 evaluates it at 3.38 in the worked domain (1.69 from machine assistance, 2.0 from sharing), turning a 1.2-year formation into a 4.05-year one.

4.7 What the Debt Weight Does and What the Constraint Does

Because debt enters the objective as a weighted term, the following diagnostic gives, for each task, the debt weight at which that term first changes the selection.

PROPOSITION 3 (Price and constraint separation). *For task i with admissible set $\mathcal{A}(i,t)$, let $\Xi(w_9)$ denote the arg max of* (27) *with the debt weight set to w_9 and $w_1, \ldots, w_8$ held in fixed ratio, which is without loss of generality because the arg max is invariant to positive rescaling of the objective. Define the threshold $w_9^\star(i)$ in Equation* (39)*:*

$$w_9^\star(i) \;=\; \inf\{\, w_9 \geq 0 \;:\; \Xi(w_9) \neq \Xi(0) \,\}, \tag{39}$$

with $w_9^\star = \infty$ when no finite weight changes the selection. Then for every $w_9 < w_9^\star(i)$ the staffed mode maximizing (27) *over $\mathcal{A}(i,t)$ is the one an objective that ignores Civic Automation Debt entirely would select.*

The proposition concerns SCV, not the dual terms the online rule adds in (29). Read as a diagnostic, it says that when $w_9 < w_9^\star(i)$ any departure from the debt-blind choice comes from the constraints of (28), not the debt weight; the objective then records the liability while a hard constraint determines the allocation, the division of labor the framework intends. We report $w_9^\star$ with every allocation. Section 6.2 computes $w_9^\star = 0.313$ for the worked case against a default of 0.22. Online enforcement of the transition and resource constraints is governed by the following accounting bound.

PROPOSITION 4 (Between-rebalance violation bound). *Suppose the online rule of* (33) *runs on a window of n arrivals, each task's outcome is observed before the next arrival is decided, and the admissibility test* (31) *is evaluated on the post-execution state using the decision-time estimates $\delta_g^{disp}(i,m,a)$. Let $\bar{\delta}_g$ be the largest amount by which execution revises one such estimate upward, and suppose Zone Z4 fallback assignments do not raise any Δ_g. Then at the end of the window the bounds in Equation* (40) *hold:*

$$\Delta_g \leq \tau_g + \bar{\delta}_g \quad \forall g, \qquad \sum_{i \in \mathcal{T}^{adm}} \sum_{m,a} x_{i,m,a}\, u_{i,m,a,r} \leq U_r \quad \forall r, \tag{40}$$

independently of n, where $\mathcal{T}^{adm}$ is the set of tasks admitted through (31)*; the resource bound is exact, and tasks routed to the Z4 fallback bypass it.*

Proposition 4 lets the just-transition and resource constraints be enforced online, with $\bar{\delta}_g = 0$ when the estimates are exact. The overshoot is at most one task's estimation error and does not grow with the window; it is not a convergence guarantee, and Section 7.3 states what one would require.

4.8 Adaptive Oversight Zones

Task classes are assigned to the four oversight zones of Table 3, which also fixes the staffed modes each zone admits.

The tiering, which formalizes the human-in-, on-, and out-of-the-loop taxonomy echoed in the EU AI Act (European Parliament and Council of the European Union, 2024), is old; what is new is its coupling to evidence through the predicates of Table 4. Promotion requires every condition *and* an affirmative panel decision, while demotion fires on any single condition, because evidence that automation works should not expand it without a political act, while evidence that it fails should contract it at once.

A zone change is recorded as a versioned policy update, and restoration after an automatic demotion always requires a panel decision.

Table 3. Adaptive Oversight Zones and the staffed modes each admits. Zone membership is evidence-conditioned by the predicates of Table 4, not a fixed sector label. ϕ^{Z3} defaults to 0.5.

Zone	Allocation principle	Admitted staffed modes
Z1 Autonomous	High-confidence, low-rights-impact tasks may be performed autonomously.	All $m \in \mathcal{M}$; any feasible roster including the null roster.
Z2 Augmented	AI or robots perform substantial work while a human remains meaningfully involved.	m with $\phi_m > 0$; any feasible roster.
Z3 Human Authority	AI may advise, but a designated human authority makes or validates the decision.	m with $\phi_m \geq \phi^{Z3}$; roster must name a competent decision-maker of record who may depart from the recommendation.
Z4 Human Reserved	Automation is restricted for reasons of dignity, democratic authority, capability preservation, or unacceptable systemic risk.	$m \in \{H,\ H{+}R\}$ with the robot as a tool under direct human control; roster must contain a competent practitioner.

Table 4. Zone transition predicates. W is a rolling window, four budget quarters by default. Promotion requires all promotion conditions plus an affirmative panel decision; demotion fires on any single demotion condition and is recorded as a versioned policy update. Thresholds are panel-set; the values shown are the defaults listed in Appendix S2 (Table S3).

	Promotion (one zone toward Z1), all required	**Demotion (one zone toward Z4), any sufficient**
1	No incident of severity 3 or above in W, and incident rate below the class baseline.	One incident of severity 3 or above, or incident rate above $1.5\times$ the class baseline.
2	Upheld-appeal rate ≤ 0.02 over W.	Upheld-appeal rate > 0.05 over W (the systemic trigger of Section 5.2).
3	Accumulated CAD for the class non-increasing over W.	Accumulated CAD for the class rising for two consecutive quarters.
4	$Res_s^{(n_s)} \geq 1.15\, R_s^{min}$ throughout W.	$Res_s^{(n_s)} < R_s^{min}$ at any evaluation.
5	$\Lambda_k \geq B_k$ in the last closed budget period.	$\Lambda_k < \bar{B}_k(t)$ for two consecutive quarters, or the infeasibility flag of Proposition 1 is raised.
6	Affirmative Weight Review Panel decision.	Input-distribution drift on the class exceeding a population stability index of 0.25, or a new *restrict* or *prohibit* rule entering the twin.

5 GOVERNANCE, CONTESTABILITY, LAWFULNESS, AND DATA PROTECTION

5.1 Policy Feedback and Weight Legitimacy

Governance is a closed loop: outcomes are evaluated by group through Equation (34), and feedback changes policy parameters only through authorized processes. The system operationalizes values set by municipal councils and the panel rather than generating policy; a parameter they have not set, such as $\theta_{k,g}$ or τ_g, is flagged as unset rather than defaulted.

The weights, budgets, access targets, group coarsening, transition ceilings, zone thresholds, and contingency parameters are set by a standing *Weight Review Panel* meeting quarterly and after any stress event. Following co-creation designs shown to raise public trust in municipal artificial

intelligence (Phothong et al., 2026; Nechesov and Ruponen, 2024), the panel couples deliberative procedures for rights- and distribution-facing parameters with technical review for operational metrics. Every panel determination is committed as a versioned update to the Policy Digital Twin, so every operational allocation traces to the configuration that authorized it. The panel includes three groups whose interests the rest of the design would otherwise leave unrepresented: organized labor, which maintains a direct stake in ϕ_m, $B_k(t)$, $\theta_{k,g}$, and τ_g (Callari et al., 2024; International Labour Organization, 2015); non-payroll workers (De Stefano, 2016; Wood et al., 2019), whom the formal headcount metric $N_k(t)$ omits and incumbent representatives have little structural incentive to defend; and the municipal legal office, which keeps rule encodings in line with statutory mandates. Residents, whose service equity enters through Eq^{srv}, take part through the deliberative procedures rather than a mandated seat; whether that suffices is for each city to judge.

The panel's authority has a limit that no choice of membership removes. Because Equation (27) is a weighted sum, the panel can reach by setting weights only *supported* allocations, those whose objective vectors lie on the boundary of the convex hull of the attainable set; since the variables are binary that set is discrete, and Pareto-optimal allocations in its non-convex regions, which may include the compromise a deliberative process would actually choose, are unreachable whatever the panel decides (Osika et al., 2023). Its authority is therefore bounded by the aggregation function rather than by its mandate, which limits the legitimacy claim. Chebyshev or ε-constraint formulations would remove the bound at some cost in interpretability, and the hard constraints of (28) already act as ε-constraints for the constructs a city judges non-negotiable.

Liability follows the zones: the decision-maker of record in Z3 and Z4, the certifying unit in Z1, and in Z2 an apportionment by the evidentiary record (Appendix S1.5).

5.2 The Contestability Procedure

Contestability in public-sector algorithms must be a first-class design requirement rather than an audit afterthought (Moon and Guha, 2026; Alfrink et al., 2023), so we specify the procedure rather than only an appeal-resolution rate. It has four elements; Appendix S1 states each in full, and Table S1 fixes the time limits.

Standing extends to affected residents and to any worker whose allocation, developmental access, or displacement is at issue, independently of contract status, with the municipal ombudsman able to act on its own motion against a pattern (the systemic-challenge route missing from most deployed systems (Vaccaro et al., 2019)). *Windows* are twenty working days to lodge and twenty to resolve, shorter with the outcome suspended where a decision is rights-affecting, and longer for worker appeals on access or displacement, which may surface only when a budget period closes.

An *evidentiary record* written by every allocation holds the roster set before and after policy filtering, each candidate's term vector and CAD decomposition, the binding constraint, the duals in force, $w_9^\star$, the configuration version, and the accountable human, and is disclosable to an appellant with standing. This is technological due process (Citron, 2008) made operational, and recording the configuration version is what makes an appeal against a *policy* possible (Almada, 2019).

The *remedy* is re-decision under Zone Z3 by a designated human who may depart from the recommendation on the record, coupled to a systemic trigger: an upheld-appeal rate above 0.05 over the rolling window demotes the class one zone by predicate 2 of Table 4 and refers it to the panel.

The evidentiary record and appeals procedure are separable from the optimizer and are the framework's most immediately deployable component: a municipality could run them on its existing allocation practice before any optimization is built.

5.3 Lawfulness of the Capability Access Constraint

A mechanism that allocates scarce developmental work against a composition target defined over gender is making a decision that equality law regulates; district of residence is not a protected ground, and a district target raises different questions. We give the analysis for EU law, and for Saudi Arabia in Section 5.4.

Directive 2006/54/EC (European Parliament and Council of the European Union, 2006) prohibits direct discrimination on grounds of sex in access to employment, vocational training, and working conditions,

and Directive 2000/78/EC (Council of the European Union, 2000) does the same for religion or belief, disability, age, and sexual orientation; both contain a positive-action provision permitting measures to prevent or compensate for disadvantages linked to a protected ground. The Court of Justice drew the boundary for sex under Directive 76/207/EEC, which Directive 2006/54/EC recast. *Kalanke* (C-450/93) held unlawful automatic, unconditional priority for women among equally qualified candidates; *Marschall* (C-409/95) upheld a similar rule with a saving clause requiring individual assessment; and *Abrahamsson* (C-407/98) held that preferring a merely sufficiently qualified candidate exceeds the exemption. *Badeck* (C-158/97) is the closest authority, because it concerned reserved *training places*, which the Court accepted where the other group could still train elsewhere. Four design features of the Capability Access Constraint are chosen against that boundary.

1. *The constraint binds an aggregate, not an assignment.* Equation (22) constrains the share of a domain's protected practice hours over a full budget period. No individual task is assigned to a practitioner because of a protected characteristic, and no practitioner is excluded from a task by the constraint.
2. *The online channel is a preference term, not an absolute priority rule.* Between rebalances the target reaches the decision through the dual $\nu_{k,g}$ in Equation (29), an incentive term in an objective where operational and safety scores retain substantial weight. It acts only on rosters that have passed the policy and safety conditions of Equation (31), but those conditions concern the task, not the individual candidate. The *Marschall* saving clause requires an objective assessment of each candidate that can override the preference, so a lawful deployment would need an individual merit floor, set administratively, that the dual cannot override. Such a floor could be encoded as a *restrict* roster guard; the framework does not supply its content.
3. *The constraint is elastic.* The slack $\varsigma_{k,g}$ in Equation (28f) turns an unattainable target into a published shortfall rather than a forced assignment, so the access constraint never requires an unlawful allocation. The other hard constraints carry no such slack and can still require particular staffing.
4. *Targets on contract status raise different questions.* The exclusion of contracted and platform-mediated practitioners, the framework's most consequential distributional concern, calls for a target on *contract status*, which, like career stage, is not a ground under either equality directive. Neither is free of legal exposure: fixed-term and part-time work are regulated by Directives 1999/70/EC and 97/81/EC, and career stage correlates with age, so a stage-based rule could be indirect age discrimination. Each needs its own analysis, but neither raises the positive-action questions of a gender target.

Two limits remain. Whether a given $\theta_{k,g}$ is proportionate is a fact-specific judgment on the evidence of under-representation, which a municipality's legal office must make and no framework can make for it; and outside the EU the analysis differs (Section 5.4). What the framework contributes is that the target, its provenance, its tolerance, and its realized shortfall are versioned and disclosable under Section 5.2, so that judgment can be made on evidence and challenged on the record. A composition-blind budget makes the same distributional choice without a recorded target, so there is no parameter to challenge.

5.4 A Second Jurisdiction: Workforce-Composition Regimes in the Gulf

Section 5.3 treats the access target as a positive-action measure justified against a prohibition. In the Gulf Cooperation Council states, composition targets over one ground, nationality, are instead an instrument of labor-market policy. We take Saudi Arabia as the instance, because its regime has been studied empirically.

Saudi civil-service appointment is confined to nationals, with non-Saudis employed temporarily where no qualified national applicant is available (Kingdom of Saudi Arabia, 1977). Private establishments, including the contractors through which municipalities procure much of their sanitation, maintenance, and inspection labor, are graded by the Nitaqat program into bands by the national share of their workforce, with thresholds varying by activity and size; the lowest band loses access to new work permits and visa renewals. Peck (2017) finds that the quota raised national employment at a substantial cost in firm exit and in reduced employment at surviving firms, borne mainly by firms with no national employees at the start.

Read against (22), three features transfer. A ministerial authority, not the allocation algorithm, sets the target; the ministry publishes and audits it; and a shortfall costs specific regulatory privileges rather than the operating license, much as the slack $\varsigma_{k,g}$ makes a shortfall costly rather than infeasible, though the program penalizes it in the objective where Nitaqat applies administrative sanctions. The proportionality inquiry of Section 5.3 does not transfer: a regulator asks of a nationality target whether it is met, not whether it may

lawfully be set. Targets on gender, district, or contract status would need their own analysis under Saudi law, which we do not attempt.

The substantive difference is what is counted. Nitaqat counts posts, so a compliant domain can staff every developmental task with an expatriate lead and form no national successor, a gap ψ_a alone does not close, since it credits developing practitioners of any nationality. With g ranging over nationality, (22) turns a headcount obligation into a formation obligation: nationals must receive the protected practice that makes them competent, not only hold posts. Where composition targets are already law, the access constraint makes an existing obligation computable and contestable.

The Just Transition Constraint meets the same structure from the other side: in a Gulf municipality the population it cannot see is large, largely migrant, and employed through the contractors Nitaqat grades (International Labour Organization and United Nations Economic and Social Commission for Western Asia, 2026), so a payroll ceiling protects the group least exposed. Extending it beyond the payroll would raise questions under the Personal Data Protection Law (Kingdom of Saudi Arabia, 2021) (Appendix S6).

5.5 Data Protection and the Civic Capability Ledger

The ledger of (10) holds per-practitioner practice totals indexed to worker groups defined over gender, contract status, and district, retained across multi-year formation periods and used to allocate work. Under the GDPR (European Parliament and Council of the European Union, 2016) that is high-risk processing, so a data protection impact assessment under Article 35 is mandatory before deployment. The design assumes four measures, set out in Appendix S10: minimization by aggregation, since the allocator needs only the career-stage test and aggregate shares; separation of protected attributes from the optimizer; small-cell suppression through the minimum cell size n^{min} of Section 3.1; and the Article 22 safeguards of the contestability procedure, whose Z3 re-decision is the required human intervention. A city that places worker-affecting task classes in Z1 should expect to defend that under Article 22, not only under the AI Act. None of this makes the ledger unproblematic; it makes its risks documented, bounded, and open to challenge.

6 A WORKED ALLOCATION: MUNICIPAL BRIDGE INSPECTION

This section carries one allocation through the framework end to end. Every number in Sections 6.2 through 6.9 is exact arithmetic on the stated inputs. The inputs are illustrative expert estimates rather than measurements; Section 6.8 maps where each conclusion survives.

6.1 The Task, the Domain, and the Candidate Set

T_i is one cycle of fatigue-crack assessment on a steel girder span, with profile $cog = 0.80$, $phy = 0.60$, $emp = 0.10$, $risk = 0.70$, $auth = 0.90$, $priv = 0.20$, $urg = 0.40$, $learn = 0.80$, $crit = 0.90$, and $d_i = 10$ hours, so by Equation (9) its developmental content is $\ell_i = 0.80 \times 10 = 8$ qualified-practice hours. A structural determination of this kind carries a statutory sign-off requirement, so under the encoding of Appendix S8.4 the Policy Digital Twin returns *restrict*, admitting only rosters that carry a licensed competent practitioner. Modes without a human slot have only the null roster, so none survives, leaving $\mathcal{M}' = \{H,\ H{+}A,\ H{+}R,\ H{+}A{+}R\}$.

The safety floor $S_i^{min} = risk_i\, crit_i = 0.70 \times 0.90 = 0.63$ removes unaided human inspection ($S = 0.55$) and human plus AI ($S = 0.60$), leaving $\{H{+}R,\ H{+}A{+}R\}$. Each surviving mode admits two rosters: a_0, a single competent licensed inspector ($\psi_{a_0} = 0$), and a_1, a competent lead with one developing inspector on an equal split ($\psi_{a_1} = 0.5$). We assume the domain's supervision rule, encoded as a further *restrict* roster guard, allows one developing practitioner per lead at this risk level ($risk_i = 0.70$), capping $\bar{\zeta}_k$ at $\phi_{H+R} \times 0.5 = 0.35$.

The domain is structural inspection of bridges and major structures: a portfolio on a routine cycle plus fracture-critical and in-depth inspections, generating $n_k = 2{,}100$ inspection events per year of duration 10 hours and learning value 0.80, so $\Phi_k = 2100 \times 8 = 16{,}800$ qualified-practice hours per year. We treat every event as an identical copy of T_i, with the same term vector, rosters, and $\ell_i = 8$; the two-mode argument

and the integer rounding of Section 6.4 rely on this homogeneity, and a heterogeneous stream could have more active modes. The workforce is $N_k = 24$ competent practitioners with annual attrition $r_k = 0.12$.

6.2 Scoring the Staffed Modes

Table 5 gives the agents' normalized estimates for the eight staffed modes, the credited developmental share $\phi_m \psi_a$, and the five debt components; the last two rows apply the default weights through (6) and (27).

Table 5. Worked bridge-inspection allocation. Normalized term valuations for the eight staffed modes, resulting Civic Automation Debt from Equation (6), and Sustainable Civic Value from Equation (27). Roster a_0 is a competent lead alone; roster a_1 adds one developing inspector at an equal work split. Shading marks the four staffed modes that survive the safety floor $S_i^{min} = 0.63$. Only D^{skill} differs between rosters here: D^{trans} is roster-indexed in Equation (6) but takes the same estimated value for both rosters, and the other three components are mode-level.

	H		**H+A**		**H+R**		**H+A+R**	
Term	a_0	a_1	a_0	a_1	a_0	a_1	a_0	a_1
Flow terms, measured on this execution window								
Q service quality	0.72	0.69	0.86	0.83	0.80	0.77	0.91	0.88
S safety	0.55	0.53	0.60	0.58	0.88	0.86	0.90	0.88
P productivity	0.40	0.35	0.62	0.57	0.70	0.65	0.88	0.83
Eq^{srv} service equity	0.70	0.70	0.72	0.72	0.70	0.70	0.74	0.74
Tr trust	0.80	0.78	0.74	0.72	0.72	0.70	0.68	0.66
$Cost$ operating cost	0.85	0.95	0.62	0.72	0.58	0.68	0.50	0.60
En energy	0.20	0.20	0.28	0.28	0.55	0.55	0.60	0.60
Pr privacy risk	0.10	0.10	0.25	0.25	0.30	0.30	0.38	0.38
Developmental shares and stock-change terms								
ϕ_m human share	1.00	1.00	0.75	0.75	0.70	0.70	0.55	0.55
ψ_a developmental eligibility	0.00	0.50	0.00	0.50	0.00	0.50	0.00	0.50
$\phi_m \psi_a$ credited share	0.000	0.500	0.000	0.375	0.000	0.350	0.000	0.275
$D^{skill} = 1 - \phi_m \psi_a$	1.00	0.50	1.00	0.625	1.00	0.650	1.00	0.725
D^{fall} fallback erosion	0.00	0.00	0.15	0.15	0.25	0.25	0.35	0.35
D^{acct} accountability dilution	0.00	0.00	0.07	0.07	0.05	0.05	0.13	0.13
D^{dep} vendor dependency	0.00	0.00	0.30	0.30	0.35	0.35	0.50	0.50
D^{trans} transition burden	0.00	0.00	0.10	0.10	0.20	0.20	0.30	0.30
$CAD_{i,m,a}$ from (6)	0.2800	0.1400	0.3816	0.2766	0.4260	0.3280	0.5004	0.4234
$SCV_{i,m,a}$ from (27)	0.2324	0.2391	0.2783	0.2773	0.3108	0.3082	**0.3426**	0.3355
Survives $S_i^{min} = 0.63$?	no	no	no	no	yes	yes	yes	yes

Read alone, Table 5 favors a fully hybrid team staffed by a lead alone: $H{+}A{+}R/a_0$ has the best quality, safety, and productivity, the lowest cost, and the highest SCV at 0.3426, despite the largest debt of the eight at 0.5004. Weighting debt does not by itself change this. With the other weights in fixed ratio, the arg max first changes at the threshold of Proposition 3,

$$w_9^\star = 0.313, \tag{41}$$

where $H{+}A{+}R/a_1$ overtakes $H{+}A{+}R/a_0$, against a default $w_9 = 0.22$. At the default, the debt weight narrows the gap between the two rosters from 0.0241 to 0.0072 objective units, closing 70% of it; on these inputs the reversal is produced by the hard constraint, not the debt weight.

6.3 The Capability Constraint Binds

Certification at inspector grade requires $h_k^{raw} = 1{,}800$ clock hours of logged practice, read as in Section 3.5, so Equation (20) gives $h_k = 0.80 \times 1800 = 1{,}440$; we target independent competence rather than statutory team-leader authority, which is a separate legal threshold, and note that h_k is the parameter

to which the example is most sensitive. With $\eta_k = 1.2$ and $B_k^{min} = 2{,}000$ hours, Equation (19) yields

$$B_k = \max\big(2000,\ 1.2 \times 24 \times 0.12 \times 1440\big) = 4{,}976.64 \text{ qualified-practice hours per year.} \tag{42}$$

The budget requires a mean credited share of at least $4976.64/16800 = 0.2962$, and condition (35) is met with about 18% to spare, at $\Phi_k \bar{\zeta}_k = 5{,}880$. The unconstrained allocation, every inspection to $H{+}A{+}R/a_0$, credits nothing and forms no successors.

6.4 Solving the Program and Reading Its Dual

Restricted to this domain and its single binding constraint, program (28) has an optimum with at most two active staffed modes, because one coupling constraint plus the simplex constraint (28b) admits at most two nonzero variables at an LP vertex. Table 6 enumerates every admissible pair reaching the required share.

Table 6. All two-mode mixes of admissible staffed modes reaching the required mean credited share $\phi_m \psi_a = 0.2962$. The enumeration of pairs is complete: no pair excluding roster a_1 can reach a positive credited share at all, and a single coupling constraint admits at most two active modes at an LP vertex. Of the single modes, $H{+}R/a_1$ alone meets the share at a mean SCV of 0.3082 and $H{+}A{+}R/a_1$ alone does not meet it. Bold marks the optimum.

Mix	Shares of tasks	Mean $\phi_m \psi_a$	Mean SCV
$H{+}R/a_1$ with $H{+}A{+}R/a_1$	28.30% / 71.70%	0.2962	**0.3277**
$H{+}R/a_1$ with $H{+}A{+}R/a_0$	84.64% / 15.36%	0.2962	0.3135
$H{+}R/a_1$ with $H{+}R/a_0$	84.64% / 15.36%	0.2962	0.3086

The optimum sends 28.30% of inspections to robot-assisted human teams and 71.70% to fully hybrid teams, and staffs a developing inspector on *every* one of them. The capability constraint's dual follows from the tie between the two active staffed modes, Equation (43):

$$0.308240 + 8\,\lambda_k(0.350) = 0.335452 + 8\,\lambda_k(0.275) \implies \lambda_k = \frac{0.027212}{0.6} = 0.045353, \tag{43}$$

with the two SCV values carried at six decimals because their difference is the whole numerator. One additional qualified-practice hour therefore costs 0.0454 objective units at the margin. This is the marginal *opportunity cost* of tightening the constraint, not the value of an hour of practice, which the framework neither estimates nor needs. Whether 0.0454 units per hour is worth paying is the panel's decision. Substituting λ_k into Equation (29) is a consistency check: the augmented scores reproduce the optimum's ranking, Equation (44):

$$\begin{aligned} \widetilde{SCV}_{H+R/a_1} = \widetilde{SCV}_{H+A+R/a_1} &= 0.435229 \\ > \widetilde{SCV}_{H+A+R/a_0} &= 0.3426 \\ > \widetilde{SCV}_{H+R/a_0} &= 0.3108. \end{aligned} \tag{44}$$

The first equality holds by construction. The binding constraint, acting through its dual, reverses the winner: the same objective, estimates, and weights now select a trainee-staffed team and move over a quarter of inspections to robot-assisted human work, so the reversal falls first on *who is on the team* and only second on *which machines are used*. Integrally, 594 tasks in $H{+}R/a_1$ miss the budget by 0.24 hours and 595 meet it, so the integer optimum's mean SCV of 0.327742 sits 7.8×10^{-6} below the LP value of 0.327750.

6.5 What Preservation Costs and What It Delivers

Unconstrained allocation yields a mean SCV of 0.3426 per task, or 719.5 objective units a year over 2,100 tasks; the constrained optimum yields 0.3277, or 688.3 units. The constraint costs 31.2 units a

year, a reduction of 4.34% (0.33–9.60% across the sweep of Section 6.8), and over the 4,976.64 hours it secures the average cost is 0.0063 units per hour, below the marginal 0.0454 as convexity requires. Table 7 decomposes which terms absorbed that cost, because an aggregate figure hides whether a city paid for its pipeline with operating cost, with quality, or with safety.

Table 7. Term-by-term decomposition of the cost of preservation in the worked domain: the unconstrained allocation (all tasks to $H{+}A{+}R/a_0$) against the constrained optimum. Cost, energy, and privacy risk are penalties in Equation (27), so a fall is an improvement for those three rows.

Term	Unconstrained	Constrained optimum	Change
Q service quality	0.9100	0.8489	-6.72%
S safety	0.9000	0.8743	-2.85%
P productivity	0.8800	0.7791	-11.47%
Eq^{srv} service equity	0.7400	0.7287	-1.53%
Tr trust	0.6800	0.6713	-1.28%
$Cost$ operating cost	0.5000	0.6226	$+24.53\%$
En energy	0.6000	0.5858	-2.36%
Pr privacy risk	0.3800	0.3574	-5.96%
CAD civic automation debt	0.5004	0.3964	-20.78%
Credited share $\phi_m\psi_a$	0.0000	0.2962	from zero
SCV	0.3426	0.3277	-4.34%

The city pays mainly in operating cost, whose normalized score rises 24.5% because every inspection carries a second person, and in productivity, down 11.5%, with smaller losses in quality and in safety, which stays above its floor. The cost figure is a change in a score rescaled against a trailing window (Section 4.3), not a share of the inspection budget; converting it to money would need a cost model the article does not supply.

Meeting the budget forms $4976.64/1440 = 3.456$ competent-equivalent practitioners a year against attrition of $24 \times 0.12 = 2.88$, a net gain of 0.576, whereas unconstrained allocation forms no successors while losing 2.88 a year. The surplus is set by the choice $\eta_k = 1.2$ through the identity $B_k/h_k = \eta_k N_k r_k$, which also makes the replenishment rate independent of h_k, although feasibility is not. What the allocation adds is that this budget is attainable from the task stream.

6.6 Pipeline Arithmetic and the Required Trainee Cohort

The optimum staffs a developing inspector on all 2,100 inspections, consuming 21,000 trainee clock hours a year, each of which accrues only $4976.64/21000 = 0.237$ qualified-practice hours, because practice is diluted by machine assistance ($\bar{\phi}_k = 0.5925$ at the optimum) and by sharing the human work with a lead ($\bar{\omega}_k = 0.5$). At $\Theta_k = 1{,}500$ domain clock hours per trainee-year, Proposition 2 gives the formation time in Equation (45):

$$\tau_k \;=\; \frac{1800}{1500} \cdot \frac{1}{0.5925 \times 0.5} \;=\; 1.20 \times 3.376 \;=\; 4.05 \text{ years}, \tag{45}$$

and the required cohort in Equation (46):

$$\hat{n}_k \;=\; 1.2 \times 24 \times 0.12 \times 4.05 \;=\; 14.0 \text{ developing practitioners}. \tag{46}$$

Both routes agree: 21,000 trainee clock hours at 1,500 per trainee-year is exactly 14.0 posts, and Little's law at 3.456 completions a year and 4.05 years' residence gives the same, an identity that lets a city compute its cohort from its task stream without modeling its pipeline. (The 14 also equals the stream's trainee capacity, but only because every inspection here carries one trainee.)

Figure 3 plots the relationship. Certification hours imply 1.2 years; under the optimal mix the same practitioner needs 4.05, 3.38 times longer (1.69 from machine assistance, 2.0 from sharing the work with a lead), so a cohort sized from certification hours would provision four trainees where fourteen are required.

A headcount plan does not show this, because the dilution lives in $\bar{\phi}_k$ and $\bar{\omega}_k$, properties of the *allocation* rather than the workforce. The result rests on reading certification hours as unshared, unassisted practice (Section 3.5); if a scheme counts hours worked beside a lead, $\bar{\omega}_k$ leaves the dilution factor, τ_k falls to 2.03 years, and $\hat{n}_k$ to 7.0.

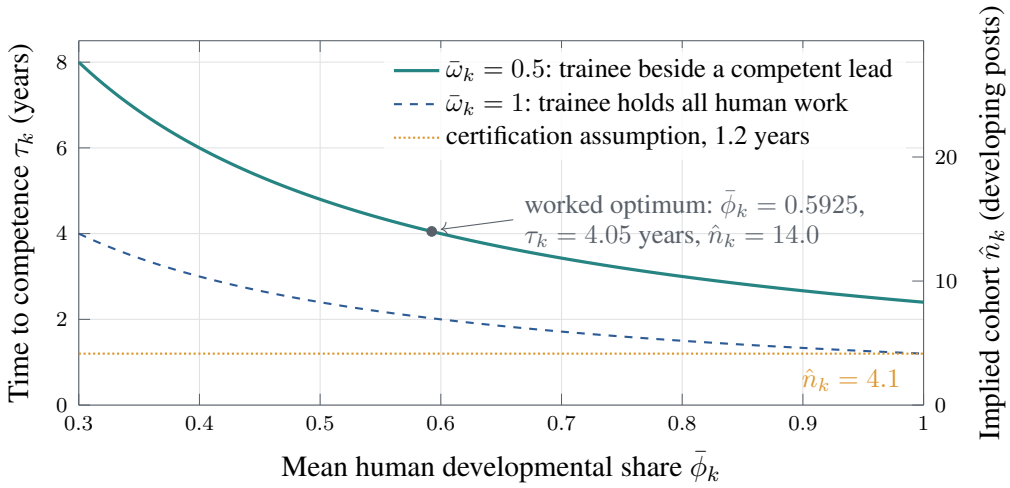


Figure 3. Time to competence under Equation (37) against the mean human developmental share of the selected staffed modes, at the worked domain's solo-practice time $h_k^{raw}/\Theta_k = 1800/1500 = 1.2$ years. The solid curve is a developing practitioner holding half of a roster's human work, as at the worked optimum of Section 6.4; the dashed curve is one holding all of it. The certification figure of 1.2 years is reached only when $\bar{\phi}_k\bar{\omega}_k = 1$, that is, unshared practice without machine assistance, the reading of certification hours adopted in Section 3.5. The right axis is the steady-state cohort of Equation (38), $\hat{n}_k = \eta_k N_k r_k \tau_k = 3.456\,\tau_k$: the certification figure implies 4.1 developing posts and the worked optimum 14.0. Every point is Equation (37) evaluated at the article's stated values; nothing is measured.

The training-economics literature suggests why an employer may not correct this dilution unprompted: skills formed on the job are largely portable, so the trainer bears the cost while others can hire the result, and training is under-provided unless frictions let the trainer recoup it (Acemoglu and Pischke, 1998; Wolter and Ryan, 2011). The dilution factor raises the trainee time a city must fund per competent practitioner by the same 3.38, a cost a training budget sized from certification hours would not show.

6.7 Who Receives the Protected Work

The 4,976.64 protected hours must go to someone, and (18) does not say to whom. Suppose the 24 incumbents are 21 men and three women, 87.5% to 12.5%, while the qualified local labor pool stands at 68% to 32%. No cell of a domain this small, not even the whole domain, meets $n^{min} = 30$, so a deployment would report its shares descriptively without a binding constraint; we set that rule aside to show what the constraint does where cells meet it. Assume a composition-blind budget recruits through the channels that produced the incumbents and so distributes protected practice in proportion, sending $0.125 \times 4976.64 = 622$ hours a year to the under-represented group, or $622/1440 = 0.432$ competent-equivalents against a domain-wide 3.456. Each replacement cycle would then reproduce the incumbent composition.

Constraint (22) intervenes on the distribution rather than the size. Setting $\theta_{k,g}$ to the labor-pool composition of 0.32 with $\varepsilon_k = 0.05$ requires $\Pi_{k,g} \geq 0.27$, or at least 1,344 hours a year, equal to 0.933 competent-equivalents, 2.16 times the composition-blind rate. The constraint changes neither B_k, the weights, nor any term estimate, but it is met only if the city also recruits developing inspectors from the under-represented group, 3.8 of the 14 developing posts of (46) rather than 1.8; that hiring decision lies outside $\mathbf{x}$. The point is not the multiplier but that who wins is set by a published parameter a panel chose and can be asked to defend. Table 8 gives the whole distribution both ways. The access constraint binds with equality because the composition-blind share of 0.125 lies below the floor of 0.27 and nothing in the

objective rewards exceeding it, so every cell is a determinate consequence of B_k and the panel's choice of $\theta_{k,g}$ and ε_k.

Table 8. Distribution of the $B_k = 4{,}976.64$ protected practice hours in structural inspection, derived by exact arithmetic. "Composition-blind" distributes protected practice in proportion to the incumbent composition; "Under (22)" applies the binding floor $\Pi_{k,g} \geq \theta_{k,g} - \varepsilon_k$ with $\theta_{k,g} = 0.32$ and $\varepsilon_k = 0.05$. Competent-equivalents are hours divided by $h_k = 1{,}440$. Groups are marginal groups on gender and contract status. No cell meets n^{min} in a domain of 24, so a deployment would report these shares descriptively; the binding floor is applied for exposition (Section 6.7). The contracted row follows from a modeling choice, not from the allocation: $\mathcal{W}_k$ holds payroll practitioners only, so non-payroll practitioners cannot be roster members and their share is zero for any $\theta_{k,g}$.

			Composition-blind			**Under** (22)		
Group	**Incumbent**	$\theta_{k,g}$	$\Pi_{k,g}$	**h/yr**	**c.eq.**	$\Pi_{k,g}$	**h/yr**	**c.eq.**
Men, permanent	0.875	0.680	0.875	4,354.56	3.024	0.730	3,632.95	2.523
Women, permanent	0.125	0.320	0.125	622.08	0.432	0.270	1,343.69	0.933
Contracted, any gender	—	unset	0.000	0.00	0.000	0.000	0.00	0.000
Total on payroll	1.000	—	1.000	4,976.64	3.456	1.000	4,976.64	3.456

One group remains uncaptured in standard payroll statistics, as the contracted row of Table 8 shows by construction. If 40% of this domain's inspection hours were delivered by contracted rope-access technicians off the payroll, they would be absent from N_k, ineligible as roster members, and given a zero share whatever $\theta_{k,g}$ is, with their displacement unrecorded. No setting of the access constraint reaches them; Sections 7.5 and 7.7 treat this as the framework's most consequential distributional limitation.

6.8 Sensitivity and the Feasibility Frontier

Every conclusion above rests on expert estimates, so Table 9 sweeps the six load-bearing parameters one at a time, reporting the optimum, the dual, the cost of preservation, and whether Proposition 1 holds, all by exact arithmetic.

The developmental staffing result holds at every feasible value swept, but its magnitude does not. No lead-only roster delivers any credit and the budget is strictly positive, so every feasible row places developing practitioners on some of the work; but across the eight feasible rows labeled "no reversal" the optimum keeps the lead-only roster on between 1.3% and 46.1% of tasks, and the dual falls from 0.0454 to between 0.0005 and 0.0033. A trainee on essentially every inspection is a property of the baseline parameterization.

The *mode* reversal is not robust. It occurs exactly where the required mean credited share B_k/Φ_k exceeds what the fully hybrid trainee roster delivers, $\phi_{H+A+R}\psi_{a_1} = 0.275$ at baseline, so it depends on the budget side (h_k^{raw}, r_k, η_k) as much as on the two estimates: lowering any of the three, or raising ϕ_{H+A+R} or ψ_{a_1}, lets the fully hybrid mode meet the budget alone and collapses the dual by an order of magnitude. That is correct behavior, since a mode that develops people should not be penalized, but the headline mode shift rests on the baseline estimates on both sides of that inequality.

Feasibility is fragile. The baseline sits at 85% of the critical value on all three axes Proposition 1 bounds, about 18% headroom on each, so raising h_k^{raw} to 2,400 hours, as a stricter licensing regime might, or attrition to 0.16, as a retirement wave could, makes the budget unattainable. The cost of preservation ranges from 0.33% to 9.60% over the values swept one at a time; joint variation could exceed that range, and the infeasible region is excluded from it. Wherever this article quotes 4.34%, that range accompanies it.

6.9 Two Parameter Shocks: Stability and Infeasibility

Two displacements are worth carrying through, because they are the two regimes Proposition 1 separates and a city could meet either. Appendix S11 gives every step.

Shock one: a retirement wave inside the feasible region. Raising attrition from 0.12 to 0.14, inside the critical rate of 0.1418, expands the budget to 5,806.08 hours and the required credited share from 0.2962

Table 9. One-at-a-time sensitivity of the worked allocation. Baseline row in bold. "Infeasible" means Proposition 1 fails, so the budget exceeds what the task stream can supply and the escalation of Section 4.6 is triggered. "No reversal" means the constrained optimum keeps the fully hybrid mode $H{+}A{+}R$ on every task and mixes its two rosters, so the roster changes on some tasks but the mode does not; "reversal" means robot-assisted human work $H{+}R$ enters the optimum.

Parameter	Value	Optimum	λ_k	Cost	Status
h_k^{raw} (clock h)	1200	28.2% $H{+}A{+}R/a_0$ + 71.8% $H{+}A{+}R/a_1$	0.0033	1.50%	no reversal
	1500	10.2% $H{+}A{+}R/a_0$ + 89.8% $H{+}A{+}R/a_1$	0.0033	1.88%	no reversal
	1800	**28.3%** $H{+}R/a_1$ **+ 71.7%** $H{+}A{+}R/a_1$	**0.0454**	**4.34%**	**baseline**
	2127	boundary of Proposition 1	—	—	critical
	2400	—	—	—	infeasible
	3000	—	—	—	infeasible
r_k (per year)	0.06	46.1% $H{+}A{+}R/a_0$ + 53.9% $H{+}A{+}R/a_1$	0.0033	1.13%	no reversal
	0.09	19.2% $H{+}A{+}R/a_0$ + 80.8% $H{+}A{+}R/a_1$	0.0033	1.69%	no reversal
	0.12	**28.3%** $H{+}R/a_1$ **+ 71.7%** $H{+}A{+}R/a_1$	**0.0454**	**4.34%**	**baseline**
	0.14	94.1% $H{+}R/a_1$ + 5.9% $H{+}A{+}R/a_1$	0.0454	9.57%	near-critical
	0.1418	boundary of Proposition 1	—	—	critical
η_k	0.8	28.2% $H{+}A{+}R/a_0$ + 71.8% $H{+}A{+}R/a_1$	0.0033	1.50%	no reversal
	1.2	**28.3%** $H{+}R/a_1$ **+ 71.7%** $H{+}A{+}R/a_1$	**0.0454**	**4.34%**	**baseline**
	1.4	94.1% $H{+}R/a_1$ + 5.9% $H{+}A{+}R/a_1$	0.0454	9.57%	near-critical
	1.6	—	—	—	infeasible
ϕ_{H+R}	0.55	—	—	—	infeasible
	0.60	84.9% $H{+}R/a_1$ + 15.1% $H{+}A{+}R/a_1$	0.1515	9.60%	reversal
	0.70	**28.3%** $H{+}R/a_1$ **+ 71.7%** $H{+}A{+}R/a_1$	**0.0454**	**4.34%**	**baseline**
	0.85	14.2% $H{+}R/a_1$ + 85.8% $H{+}A{+}R/a_1$	0.0188	3.02%	reversal
ϕ_{H+A+R}	0.45	57.0% $H{+}R/a_1$ + 43.0% $H{+}A{+}R/a_1$	0.0241	7.00%	reversal
	0.55	**28.3%** $H{+}R/a_1$ **+ 71.7%** $H{+}A{+}R/a_1$	**0.0454**	**4.34%**	**baseline**
	0.60	1.3% $H{+}A{+}R/a_0$ + 98.7% $H{+}A{+}R/a_1$	0.0023	1.62%	no reversal
	0.70	15.4% $H{+}A{+}R/a_0$ + 84.6% $H{+}A{+}R/a_1$	0.0009	0.63%	no reversal
ψ_{a_1} (supervision)	0.40	—	—	—	infeasible
	0.4232	boundary of Proposition 1	—	—	critical
	0.50	**28.3%** $H{+}R/a_1$ **+ 71.7%** $H{+}A{+}R/a_1$	**0.0454**	**4.34%**	**baseline**
	0.67	19.6% $H{+}A{+}R/a_0$ + 80.4% $H{+}A{+}R/a_1$	0.0005	0.33%	no reversal

to 0.3456. The same pair of modes still brackets the requirement, so λ_k stays at 0.0454 and only the mix moves: robot-assisted human work rises from 28.3% to 94.1%, the cost of preservation from 4.34% to 9.57%, and the domain operates at 98.7% of its ceiling. This is the stabilizing direction the capability feedback is designed to have. The arithmetic does *not* establish that an implementation updating duals at a finite rebalance cadence converges to this point rather than oscillating around it (Section 7.3).

Shock two: a retirement wave outside it. At attrition 0.25 the budget becomes 10,368 hours against the ceiling of 5,880, a deficit of 4,488 hours or 3.12 competent-equivalents a year that *no* allocation of this task stream can form. Of the restoration options of Section 4.6, two developing practitioners per lead leaves a deficit of 2,528 hours; 3,703 supervised inspections a year closes it; and lowering η_k to 0.68 closes it by deciding not to replace the full cadre, a contraction that should be recorded as one. The binding quantity is $\Phi_k \bar{\zeta}_k$, the practice the task stream can carry, not recruitment: a municipality could recruit thirty trainees and still fail, because with one trainee per inspection the stream supports at most $21{,}000/1{,}500 = 14$ trainee posts at the prevailing automation level.

7 DISCUSSION

7.1 An Analytic Model of Debt Accumulation

The debt components of (6) are per-task quantities; what a city experiences is their accumulation, and its shape distinguishes a strategy that replenishes a stock from one that does not. Let $p_j(t) \in [0, 1]$ be the fraction of component j's required replenishment delivered at time t, let debt accrue in proportion to the unmet fraction, and let each component approach its own asymptote with its own time constant,

Equation (47):

$$\dot{C}_j(t) = c_j\,[1 - p_j(t)], \qquad p_j(t) = p_{j,\infty}\big(1 - e^{-t/\tau_j}\big). \tag{47}$$

Integrating from $p_j(0) = 0$ and weighting by the debt weights of (6) gives the total debt trajectory in Equation (48):

$$C(t) = \sum_j w_j\, c_j \Big[(1 - p_{j,\infty})\, t \;+\; p_{j,\infty}\, \tau_j \left(1 - e^{-t/\tau_j}\right)\Big], \qquad w_j \in \{\alpha, \beta, \gamma, \delta, \epsilon\}. \tag{48}$$

Modeling all five components rather than D^{skill} alone matters because different mechanisms drive each $p_{j,\infty}$: the hard constraint (18) drives skill formation to one, with τ_{skill} set, by assumption, equal to the formation time of 4.05 years from Proposition 2; the reserve (26) drives fallback to one; revisable zones and (23) drive accountability and transition high; and *nothing in the framework drives vendor dependency*, which keeps accruing linearly. Table 10 fixes every parameter, and Figure 4 plots the result.

Table 10. Per-component replenishment parameters for Equation (48), one row per strategy, reported as $p_{j,\infty}$ with τ_j in years in parentheses. The accrual rate is $c_j = 10$ index units per year for every component of every strategy, except that Cost/Performance carries $c_{acct} = 14$ and $c_{dep} = 16$. These are the article's illustrative choices, not measured quantities, and the rows for the two external baselines, which are not implemented, encode our reading of their mechanisms; they are tabulated so that Figure 4 is reproducible by direct substitution rather than asserted. $\tau_{skill} = 4.05$ is set equal to the formation time of Proposition 2. The final two columns are computed from the row, not fitted to the plot.

Strategy	$p_{skill,\infty}$ (τ)	$p_{fall,\infty}$ (τ)	$p_{acct,\infty}$ (τ)	$p_{dep,\infty}$ (τ)	$p_{trans,\infty}$ (τ)	**Residual slope**	$C(10)$
Automation-First	0	0	0	0	0	10.00	100.00
Cost/Performance	0	0	0	0	0	11.44	114.40
Capability Matching	0.35 (4.05)	0.30 (2.0)	0	0	0	8.36	88.54
Ergonomics-Aware Role Allocation	0.40 (4.05)	0.45 (2.0)	0	0	0	7.89	85.02
Human-First	0.15 (4.05)	1.00 (0.5)	1.00 (0.5)	1.00 (0.5)	1.00 (0.5)	2.38	28.96
CivicWorkOS	**1.00 (4.05)**	**1.00 (1.0)**	**0.85 (1.5)**	**0.10 (2.0)**	**0.95 (1.0)**	**1.45**	**31.51**

No strategy here bounds debt, CivicWorkOS included. Its saturating components contribute a bounded 17.98 index units, 11.34 of them from skill formation, but the residual accrues at 1.45 units a year without limit, 1.08 of it vendor dependency. A capability budget converts *skill* debt from unbounded to bounded and says nothing about the other components except through outvotable weights. Vendor dependency is a procurement variable the allocation mechanism does not control; it belongs to the governance of procurement, where the specification of an algorithmic system is itself a policy decision requiring administrative process (Mulligan and Bamberger, 2019), as reskilling coverage is (Section 3.6).

Human-First beats CivicWorkOS on this metric for thirteen years. Keeping humans on everything replenishes the other four components at once, while CivicWorkOS fills its pipeline over 4.05 years. The curves cross at $t = 13.23$, both at 36.73 index units, and by year twenty stand at 46.89 against 52.89, because Table 10 gives Human-First a skill replenishment of only 0.15, on the reasoning that maximizing human share with availability-ordered rosters does not place developing practitioners on the work. On a ten-year horizon and this index alone a city should prefer Human-First, paying in operational terms the index omits; debt is one construct among thirteen, not the objective.

The ordering of the curves is not evidence. It follows from the replenishment parameters of Table 10, which encode our reading of what each strategy's mechanisms would deliver; the crossover at 13.23 years is a consequence of those inputs, and Figure 4 describes the debt accounting under them, not cities. Whether $p_{skill,\infty}$ can reach one across heterogeneous task streams without excessive Zone Z4 fallback is what the protocol of Section 7.4 would answer.

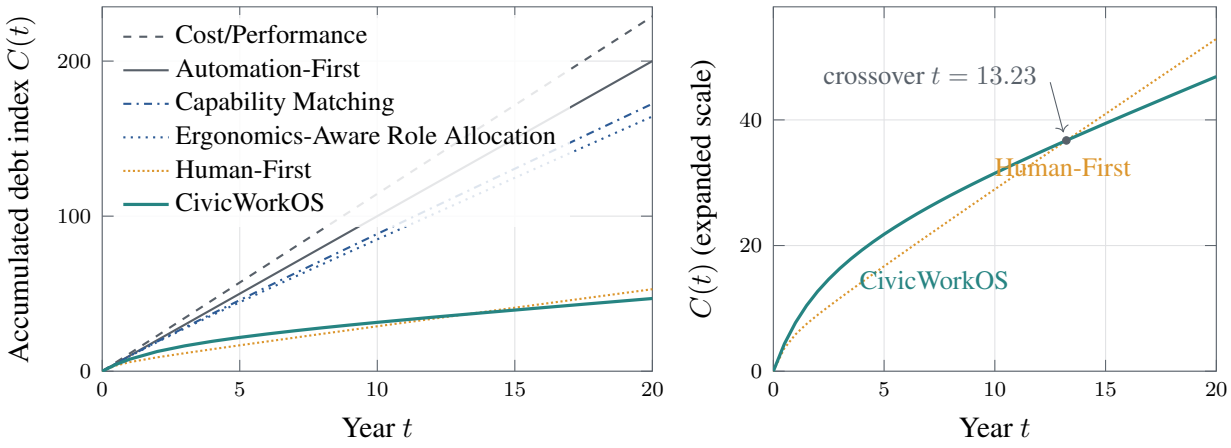


Figure 4. Accumulated Civic Automation Debt under Equation (48) at the per-component parameters of Table 10. Every plotted point is that equation evaluated at those parameters; nothing here is measured or simulated. *Left:* all six strategies over twenty years. Automation-First is normalized to 100 at year ten by setting $c_j = 10$ for every component, and Cost/Performance exceeds it because it carries elevated accountability and dependency accrual. *Right:* the two capability-forming strategies at an expanded scale. CivicWorkOS drives skill formation and fallback to full replenishment through hard constraints, with $\tau_{skill} = 4.05$ years set equal to the formation time of Proposition 2, but has no mechanism for vendor dependency, so its curve does not saturate. Human-First accumulates less debt than CivicWorkOS until year 13.23 and more thereafter, because its residual slope of 2.38 index units per year, all of it unformed skill, exceeds CivicWorkOS's 1.45, three-quarters of which is vendor dependency.

7.2 What the Framework Establishes, and What It Does Not

Section 6 establishes by arithmetic that the constraint system is computable on an illustrative domain, that its constraints bind, that the dual of the binding constraint is legible, and that with (22) the distribution of protected practice is set by a published parameter rather than by recruitment; Section 6.8 maps where those results survive. None of that shows any input estimate is correct: the estimates are the authors' judgment, the elicitation protocol of Appendix S5 has not been run, and sensitivity analysis is a weaker substitute for measurement.

Against the research questions: RQ1 and RQ5 are answered by specification (Sections 3 to 5); RQ2 by construction, through the flow–stock separation of Section 3.1, though whether the debt components measure the losses accurately is untested; RQ3 and RQ4 analytically for one illustrative domain (Sections 4.6 and 6.3 to 6.7); and RQ6 and RQ7 are specified, but only the unexecuted protocol of Section 7.4 could answer them.

Nor is a structural property of the optimization model an empirical finding. Whenever a capability constraint binds, the constrained optimum's SCV can be no higher than the unconstrained one, so preservation has a non-negative cost in objective units; that sign is structural. Which terms absorb the cost, and how large it is, depend on the inputs. In the worked domain it falls mainly on operating cost and productivity (Table 7) and is 4.34%, ranging from 0.33% to 9.60% over the one-at-a-time sweep of Section 6.8. A single-domain sweep cannot say whether the system remains satisfiable at municipal scale.

7.3 What a Guarantee for the Online Rule Would Require

Proposition 4 says nothing about how the online rule performs against the program it approximates. The bound to seek is regret of order $O(\sqrt{T})$ in the number of arrivals T against the offline optimum of (28), with cumulative violation of (28e) and (28f) of the same order. Close relatives in online allocation and constrained MDPs have such bounds (Devanur and Hayes, 2009; Balseiro et al., 2023; Mehta et al., 2007; Altman, 1999; Paternain et al., 2019), and the small-request condition they need holds here: one inspection carries at most 2.8 credited hours against a budget of 4,976.64. Four features keep them from applying directly: the duals are refreshed only at rebalances; the capability budget is a covering constraint, protected

only by reserving future capacity, for which the catch-up threshold (32) is an unanalyzed heuristic; its coefficients change as assignments move practitioners across the competence threshold; and the linearized access constraint has coefficients of both signs over a combinatorial roster space. Appendix S9.4 states each in full. The second and third are specific to capability formation, and are where we expect the difficulty to lie.

7.4 A Specified Evaluation Protocol

The protocol summarized here *has not been executed*; Appendix S2 specifies it in full, in advance, so that it can be audited against a plan. It is a discrete-event testbed over six municipal sectors and 803,100 tasks a year, run for ten years so that retirement waves and pipeline decay fall inside the horizon, comparing Automation-First, Cost/Performance, Human-First, and CivicWorkOS (all configurations of Equation (27)) with two external baselines, Capability Matching after Ranz et al. (2017) and Ergonomics-Aware Role Allocation after Merlo et al. (2023), under paired Wilcoxon tests with Holm–Bonferroni control over sixty tests and seven stress scenarios, two of which Section 6.9 works out by exact arithmetic.

With four of six comparators configurations of the proposed objective, the evaluation can show that the mechanism does what it was designed to do, not that it outperforms an independent alternative; the comparison is incomplete until the two external baselines, neither yet implemented, are run. The open-data portals the protocol calibrates against also belong to high-income cities with few non-payroll workers, so the evaluation would understate the population the framework fails, a bias in the framework's favor.

The ten criteria (eight hypotheses and two calibrated benchmarks) are *pre-specified*, not pre-registered; they will be lodged on the Open Science Framework before execution. Their thresholds were set with the worked example's arithmetic in hand, a weaker form of pre-specification than setting them blind.

7.5 Who Wins and Who Loses

The intended winners are residents whose critical services retain human fallback: if the reserve is sized and maintained as specified, constraint (26) turns an AI-service outage into a degradation rather than an interruption, though a reserve sized for an outage is not sized for a surge.

Workers who receive protected developmental work gain, provided $\theta_{k,g}$ is deliberately calibrated rather than inherited. A capability budget rations a scarce good, and under a composition-blind budget the under-represented group of Section 6.7 receives 622 hours a year against 1,344 under the access constraint.

One group is served worst. Workers outside the municipal payroll (contracted, agency, informal, or platform-mediated (De Stefano, 2016; Wood et al., 2019)) are absent from $N_k(t)$, ineligible as roster members, and unprotected by Equation (23). Where they perform much of a city's sanitation, inspection, and logistics labor, CivicWorkOS as specified would preserve the permanent workforce while the contracted one absorbed the displacement, and report a satisfied access constraint while doing so; Table 8 prints the zero row, and Section 5.4 explains why it is large in a Gulf city. No parameter setting fixes this, because it follows from a data substrate that presumes formal employment.

The framework also legitimates automation as much as it constrains it: a documented SCV computation, a satisfied budget, and a versioned panel decision can justify displacing workers once the budget is met. Whether that is a feature or a hazard depends on who sets the weights, which is why panel composition is a design parameter and why Proposition 3 lets a city see whether its objective or its constraints are deciding. The framework is not neutral between contested outcomes; it makes a political choice explicit and traceable.

7.6 Threats to Validity

Construct validity. Qualified-practice hours are a proxy, and hours are not learning; without the retention functions of the skill-decay literature (Arthur et al., 1998; Tatel and Ackerman, 2025), the ledger overstates competence under intermittent practice. *Internal validity.* Most terms are reported by agents with an interest in their value, and ϕ_m, which enters four channels, comes from the Workforce Development Agent (Appendix S8.8), whose remit favors a larger budget; cross-validation against observed outcomes is only a partial remedy. *External validity.* A single modeled city of one size and a high-income calibration base say nothing about a majority-informal workforce, where the problem is most severe.

7.7 Limitations

- *No measured inputs.* The estimates and parameters are the authors' own; the elicitation protocol that would replace them (Appendix S5) has not been run.
- *Weight legitimacy.* The quarterly panel of Section 5.1 requires deliberative processes a municipality would have to build, and some parameters move faster than any panel meets.
- *Linear scalarization.* The panel can reach only supported allocations (Section 5.1).
- *No guarantee for the online rule.* Beyond Proposition 4 there is no regret or cumulative-violation bound (Section 7.3); safe reinforcement learning (García and Fernández, 2015) offers a further route.
- *Unmeasured tractability.* Program (28) is NP-hard in general, and no claim here depends on where it stops being tractable.
- *No envy guarantee.* The access constraint is a share constraint, not a fair-division mechanism (Bouveret et al., 2016), so members of one group can receive very different practice.
- *Privacy and gaming.* The measures of Section 5.5 bound but do not remove the ledger's exposure, and ϕ_m is the most gameable quantity in the framework.
- *Machine-checkability and jurisdiction.* Decidable predicates cannot capture the discretionary judgment that constitutes much of a public service (Lipsky, 2010), and Sections 5.3 and 5.4 cover two jurisdictions without amounting to legal advice in either.
- *Formal-employment presumption.* The framework preserves the visible payroll workforce and cannot see the rest. This is the limitation we would fix first.

8 CONCLUSIONS

An allocation decides more than whether humans, agents, or robots perform a task better: it changes what a city will be able to do, and who will be able to do it. CivicWorkOS governs these decisions jointly rather than one task at a time.

Its central claim is that human expertise, fallback competence, and accountability are renewable but depletable civic resources, and that automation consuming them without replenishment creates Civic Automation Debt. The concern has established lineages (Bainbridge, 1983; Cunningham, 1992; Burke et al., 2004); the contribution is a decision variable that names who performs the work, a capability constraint that credits only the people it is sized to form, an access constraint that makes their distribution a contestable choice, and a necessary feasibility condition with the actions to take when it fails.

On illustrative inputs, the worked bridge-inspection allocation shows the following. Meeting the capability budget reverses the winning staffed mode, at a marginal opportunity cost of 0.0454 objective units per protected-practice hour and a cost of 4.34% of annual objective value (0.33–9.60% across the sweep). At the chosen replacement factor of 1.2 it forms 3.46 competent-equivalents a year against attrition of 2.88, where unconstrained allocation forms nobody. Given the recruitment it presupposes, an access constraint more than doubles the protected practice reaching an under-represented group, from 622 to 1,344 hours a year. And machine assistance (a factor of 1.69) and sharing the work with a lead (2.0) stretch time to competence by 3.38, from 1.2 years to 4.05, so a plan sized on certification hours provisions four trainee posts where fourteen are required (Figure 3).

Three limits bear restating. The framework bounds skill debt but not vendor-dependency debt (Figure 4). Its feasibility margin in the worked domain is about 18% on three axes, so a stricter certification regime or an older workforce reaches the boundary. And it cannot see workers off the payroll, whom the exposure literature identifies as most exposed.

Future work should measure the inputs through the elicitation protocol; prove regret and violation bounds for the online rule (Section 7.3); extend the ledger to contracted and platform-mediated practitioners; let c_w depreciate under intermittent practice; and pilot the evidentiary record and contestability procedure of Section 5.2 against an existing municipal allocation practice, which a city could adopt without the optimizer.

ABBREVIATIONS

AI, artificial intelligence; CAD, Civic Automation Debt; EU, European Union; GDPR, General Data Protection Regulation; HCPB, Human Capability Preservation Budget; ILO, International Labour Organization; ISO, International Organization for Standardization; LP, linear program; MDP, Markov decision process; SCV, Sustainable Civic Value.

ETHICS STATEMENT

This study involved no human participants, no animal subjects, and no personal data. All parameter values reported are the authors' structured estimates, and all results are analytical. The expert-elicitation protocol specified in Appendix S5 has not been executed; its execution would require institutional ethics approval and written informed consent from participating practitioners, as Appendix S5.4 states. Ethical review and approval were therefore not required for this work in accordance with institutional and national requirements.

DATA AVAILABILITY STATEMENT

No new datasets were created or analyzed in this study. Every parameter value and every result of Sections 6 and 7.1 is specified in this article and reproducible from the reported values by direct arithmetic. A reference implementation that recomputes every checkable number in this article, check by check, is available at `https://github.com/aliakarma/CivicWorkOS`. Code under `sim/` in that repository runs on clearly labeled synthetic demonstration data; it exercises the software shape of the protocol specified in Section 7.4 and Appendix S2, and it must not be cited as empirical municipal validation.

AUTHOR CONTRIBUTIONS

TAS and AAk conceptualized the study and developed the research methodology. RMA supervised the study and provided guidance on the research design and interpretation of the findings. MTN and AAl contributed to the investigation, formal analysis, and validation. TAS and AAk developed the reference implementation and its verification, and prepared the visualizations. TAS prepared the initial manuscript. All authors contributed to the revision and editing of the manuscript, critically reviewed the final version, and approved the submitted version. AAk denotes Ali Akarma and AAl Abdulaziz Alqurashi.

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CONFLICT OF INTEREST

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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